Convolutional Neural Networks For Tissue Screening In Intestinal Immunohistochemistry Images

submitted by

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for the degree of Master of Science

of the

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September 2026

Summary

The Human Protein Atlas contains millions of immunohistochemistry images, where the subcellular localisation of proteins provides insight into their function. Systematically cataloguing these positions across the proteome could enable novel biological inquiries. However, image analysis cannot proceed until an expert determines whether the tissue section is readable, a judgement that has not previously been documented in the atlas.

This dissertation formalises the readability judgement in collaboration with a domain expert, focusing specifically on small-intestine tissues. A usability rubric was developed with the domain expert. Then, an annotation tool was used by the domain expert to output an expert-labelled dataset. Furthermore, a convolutional classifier is trained and evaluated using a grouped protocol designed to elucidate training limitations. Additionally, a localisation model is proposed to identify readable regions suitable for histological examination. Statistical analysis was used to gain a thorough understanding of the advantages and limitations of the finalist models that were selected for screening. The resulting datasets are a unique expert-labelled usability resource for intestinal immunohistochemistry. The experiments further showed that expert label quality, rather than data volume, determined what such screening could achieve.

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Acknowledgements

I am grateful to the following people for their support during this project. My supervisor, Neil Langmead, who joined the project at short notice and advised me throughout. Dr Randy Mrsny, who guided me through the biological obstacles the project raised and gave his time to create a dataset that did not exist before. My friends Zaina Alhemeiri, Alex Lozano and Atef Khasawneh, who supported me throughout.

Use Of Generative AI

I acknowledge that this dissertation is my own work. This assessment is classified as Type B, and generative AI tools were used as assistive tools only, for figures, structural planning for code and the dissertation, as follows. The figures that were created by Claude (Anthropic, `https://claude.ai`) were Figures 2-1, 2-2, 2-3, 3-1, 3-2, 3-3, 3-4, 3-5, 4-3, 4-4, 4-5, 5-1, 5-2, 5-3, 5-4, 5-5, 6-1, 6-2 and A-1.

Gemini 2.5, in its Flash and Pro variants (Google, `https://gemini.google.com`), was not used as a writing aid. It forms part of the method, providing the seven-member labelling panel that produced the silver dataset, as described in Chapter 4.

All prompts were written by me; all outputs were checked against the source material and the code, and I take full responsibility for the content, analysis and conclusions of this dissertation.

Chapter 1

Introduction

1.1 The Human Protein Atlas As A Spatial Map Of The Proteome

The Human Protein Atlas (HPA) is a large biomedical research programme, launched in Sweden, that maps where proteins are found across the body's cells, organs and tissues (Uhlén et al. 2015). It classifies the expression of the roughly 20,000 human protein-coding genes across all major organs and tissue types (The Human Protein Atlas 2019), with antibody-based imaging available for more than 15,000 of these genes across 45 normal tissue types (The Human Protein Atlas 2026b). The atlas consists largely of images and is one of the largest open-access collections of tissue imagery. The atlas includes more than ten million images generated using two staining techniques, immunofluorescence (IF) and immunohistochemistry (IHC). Both stain a chosen protein within a tissue section so its location can be visualised under a microscope (Thul and Lindskog 2018). This resource underpins research across systems biology, cancer biology and protein localisation (Lindskog 2016).

1.2 Applications And Imaging Challenges In Protein Localisation Analysis

HPA imaging reveals which proteins are present and where they localise, exposing how expression differs from tissue to tissue (Thul, Åkesson, et al. 2017). A single protein can act in several pathways, adopting distinct roles according to where it localises in the cell. Mapping localisation across the entire proteome allows differences among thousands of proteins to be measured simultaneously.

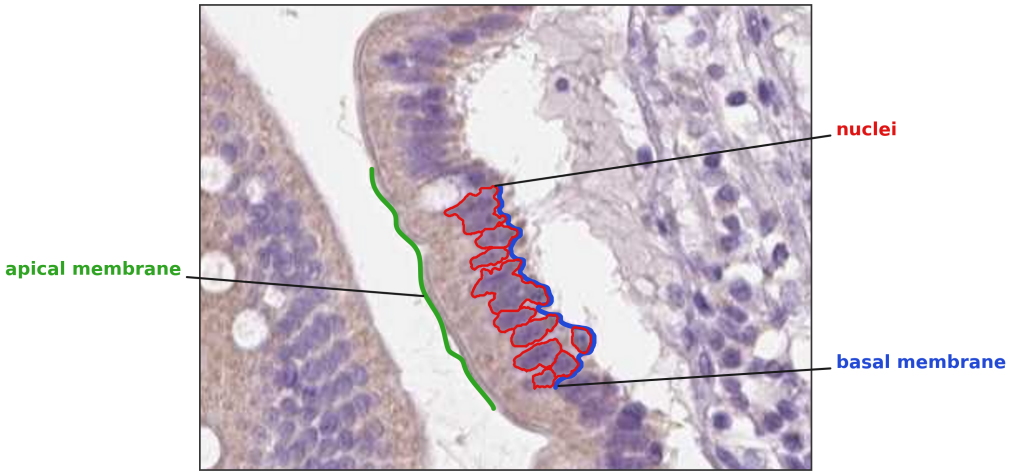

Figure 1-1: The apical-to-basal axis in intestinal epithelium. The base image is from the Human Protein Atlas (Uhlén et al. 2015); the axis annotation was added for this project. The apical membrane (green) faces the gut lumen, the basal membrane (blue) faces the connective tissue beneath, and the epithelial nuclei (red) lie between them.

The gut makes this especially vivid. It is lined by a sheet of epithelial cells. The surface of each cell facing the gut lumen differs from the surface facing the underlying tissue, a property known as cell polarity. The apical face points toward the lumen, where food passes, as shown in Figure 1-1. It facilitates absorption and secretion, aided by a dense fringe of microvilli called the brush border that runs along it. In contrast, the basal face rests on the underlying connective tissue and handles adhesion and signalling. The lateral face abuts its neighbour, sealed near the apex by tight junctions illustrated in Figure 1-1. As the apical and basal surfaces discharge contrasting functions (Buckley and St Johnston 2022), a protein's position on either surface likely dictates its role (Thul, Åkesson, et al.

2017). In the intestine, that position is often the most informative detail the atlas records about a protein.

Whether the protein's compartment can be read from an image often depends on how the tissue is cut. The apical-to-basal axis is only visible when the epithelium is cut roughly at right angles to the cell sheet. When it is, the apical surface, nuclei, and basal surface line up in order. However, an oblique cut scrambles the axis; the two faces can no longer be teased apart, and the section is unreadable. The two-stage pipeline handles this. The first stage decides whether an image contains any tissue with a clean axis; the second locates those regions.

The binding constraint is sheer volume. Annotating these images by hand is estimated to have consumed twenty-six pathologists some seven and a half years across roughly sixteen million images (Navani 2016). Not enough expert reviewers can label images by hand, as the volume of new images outpaces the rate at which they can be annotated. Therefore, researchers have turned to scalable computer-based methods that use deep learning to process large numbers of histological images.

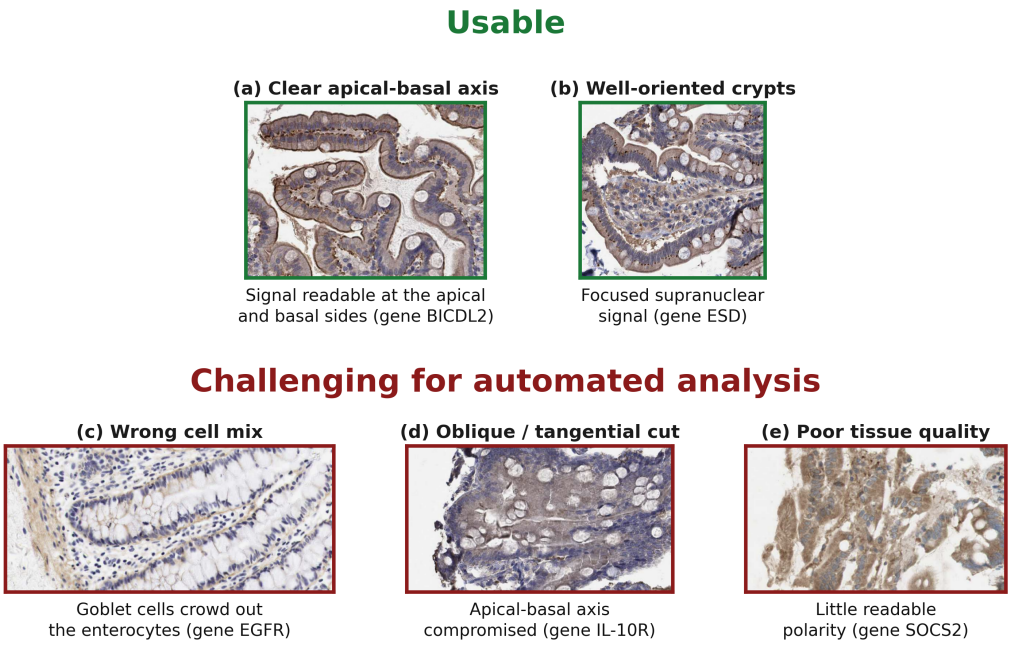

Figure 1-2: Usable versus challenging tissue in HPA small-intestine images: readable, well-oriented epithelium against sections spoiled by folds, oblique cuts or non-epithelial tissue. All panels are Human Protein Atlas images (Uhlén et al. 2015), selected and labelled for this project.

Two obstacles beset automated analysis here. The first is image quality. Each image is a high-resolution scan of a stained tissue section, and preparing that section is imperfect and often introduces artefacts. An artefact is a structure not present in the living tissue but that appears during preparation (Taqi et al. 2018). Examples include section folding over on itself, uneven stain spreading, and tissue cut at an angle that could distort its structure. At worst, the region a reviewer needs is absent from the image altogether. These artefacts hide the architecture the analysis depends on, as shown in Figure 1-2. Automated correction is unreliable, so affected images are flagged rather than salvaged (Lindskog 2016). The second is pinpointing tissue worth advancing to analysis. Even with usable images, the readable regions can be small and scattered, embedded in unanalysable tissue such as muscle and connective tissue. The two stages tackle these problems in turn, quality first and then localisation.

1.3 A Two-Stage CNN Pipeline For Slice Quality And Region Localisation

Convolutional Neural Networks (CNNs) are widely used for histological image analysis. Earlier rule-based methods required patterns to be specified by hand, but the variability of tissue samples makes those patterns hard to capture with fixed rules (Srinidhi et al. 2021). CNNs avoid this by learning patterns directly from training data (Yamashita et al. 2018). They have proven reliable across a range of medical imaging applications, including classification and segmentation of histological slides (Litjens et al. 2017).

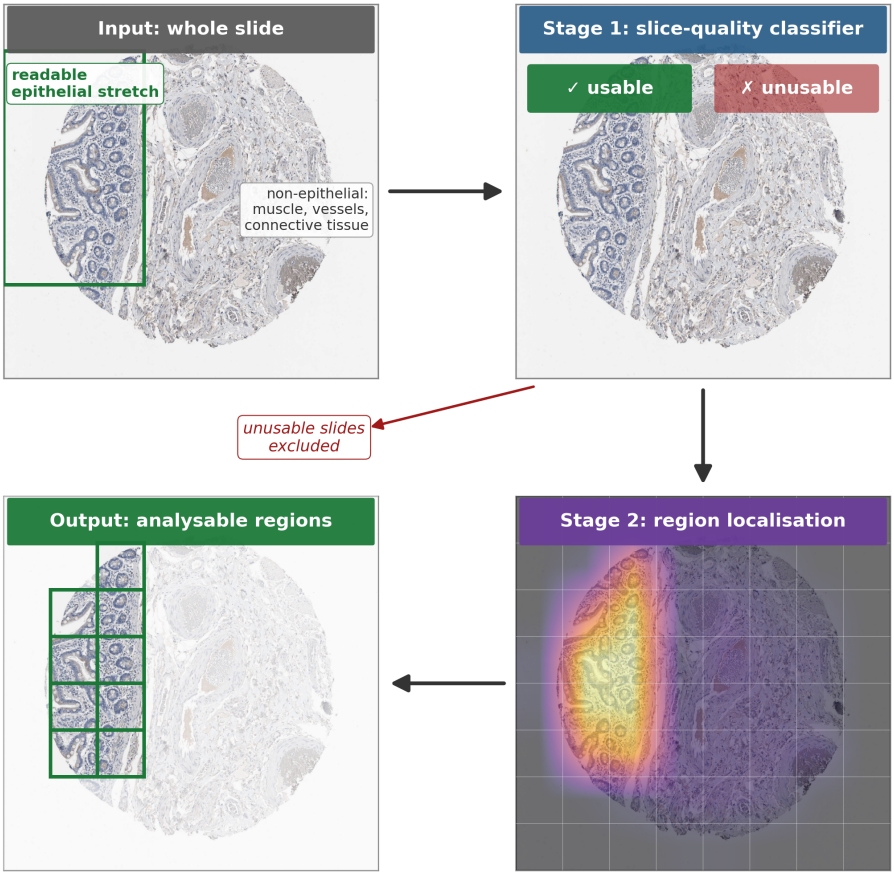

Figure 1-3: The two-stage pipeline applied to a core that is mostly muscle, vessels and connective tissue, with one readable epithelial stretch on the left. Base image from the Human Protein Atlas (Uhlén et al. 2015); pipeline annotation added for this project.

This project uses CNNs in a two-stage pipeline, as depicted in Figure 1-3. Stage 1 is a slice-quality classifier that determines whether an image contains analysable tissue and filters out images that do not. Stage 2 then localises where the tissue is analysable within a usable image, highlighting the readable regions a reviewer should examine. Both stages are necessary, since an image can merit reading even when most of it is muscle or connective tissue rather than epithelium. An image is usable if it contains at least one readable region. Figure 1-3 illustrates a difficult example. Although almost the entire core is non-epithelial, a single readable stretch is enough for Stage 1 to pass the image; Stage 2 then flags that readable stretch for the reviewer.

1.4 The HPA Small-Intestine Subset

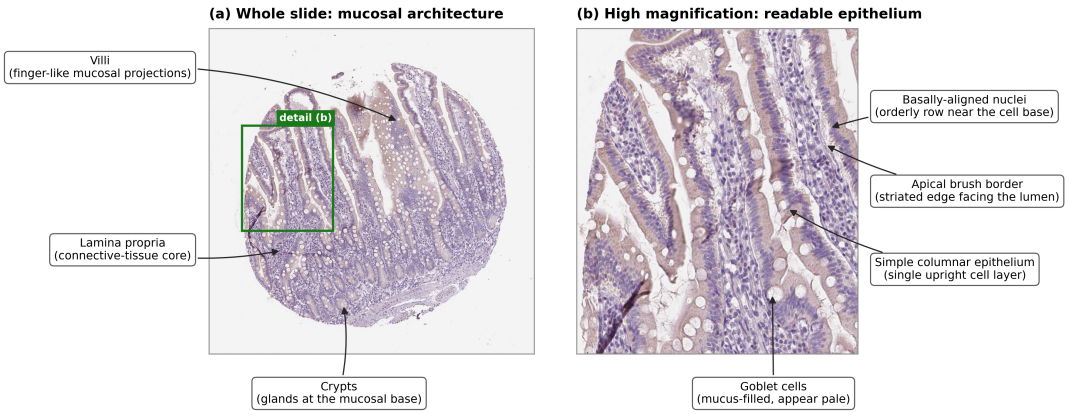

Figure 1-4: A usable duodenum image, the whole-image unit the classifier judges, with the full core in panel (a) and a high-magnification detail of readable epithelium in panel (b). The base image is from the Human Protein Atlas (Uhlén et al. 2015); the annotations were added for this project.

This project uses the small-intestine subset of the Human Protein Atlas Tissue Atlas (Uhlén et al. 2015), the largest open-access immunohistochemistry corpus. The images are shared under a Creative Commons licence (CC BY-SA) that permits academic reuse, provided the source is credited, and anything built from it is shared on the same terms. The small intestine is one of four gastrointestinal tissues the HPA roadmap prioritises (Digre and Lindskog 2023), so the project

follows that lead. Each image represents a single tissue core, which is a small circular sample extracted from a microarray slide, as illustrated in Figure 1-4.

1.5 Project Aims And Objectives

The project aims to deliver a two-stage pipeline in which a slice-quality classifier judges each tissue image as a whole and a localisation stage recovers its readable epithelium. The slice-quality classifier is the primary contribution. Five objectives follow. The first is to define usability with a domain expert and to encode that judgement in a labelling rubric. The second is to compare several candidate backbones empirically rather than assuming that any transfers to IHC. The third is to extend the expert's labels with model-guided mining and machine annotation, testing how far imperfect labels can substitute for scarce expert ones. The fourth is to evaluate every model under donor-grouped cross-validation with per-class metrics, so that reported performance reflects unseen donors. The fifth is to localise the analysable tissue from image-level labels alone. The slice-quality classifier targets a macro-F1 (the mean of the per-class F1 scores) of at least 0.85. This sits above every measured baseline yet below the domain expert's own repeat consistency.

Chapter 2

Literature Review

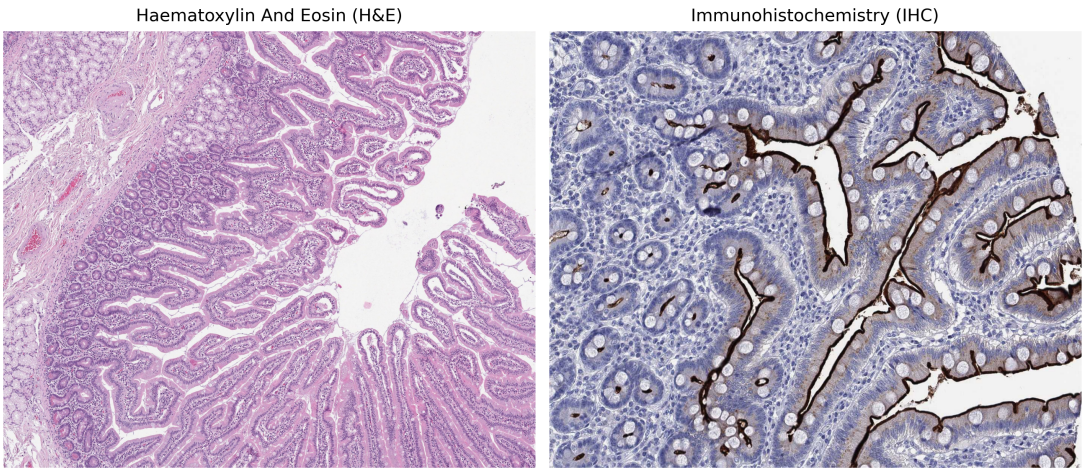

Figure 2-1: The same type of small-intestine tissue under the two stains, with H&E showing overall structure and IHC marking a single protein, ACE2. Both images are from the Human Protein Atlas (Uhlén et al. 2015).

This chapter reviews the literature informing the two-stage pipeline. The images under analysis are generated using immunohistochemistry (IHC). In contrast, most computational pathology research utilises haematoxylin and eosin (H&E), a general-purpose stain that visualises overall tissue structure rather than individual proteins, as shown in Figure 2-1. Consequently, the methods discussed in this chapter were primarily developed for a different image category.

2.1 The Annotation Bottleneck In The Human Protein Atlas

Annotating the atlas presents a substantial manual challenge, as it contains over thirteen million IHC images (Uhlén et al. 2015). Each image is stained for a single protein and then reviewed by a pathologist (Uhlén et al. 2015). Despite this rigorous process, annotation cannot keep pace with the collection's rapid expansion. The atlas has already characterised the small intestine in detail (Gremel et al. 2015).

The Human Protein Atlas (HPA) has progressed from championing automation (Lindskog 2016) to enacting it. Sullivan et al. (2018) addressed the shortage of labelled data through crowdsourcing, recruiting a large non-expert community. Specifically, 322,000 players of the online game EVE Online sorted protein-localisation images. Together, they produced approximately 33 million labels. Performance was measured with the F1 score, which balances precision and recall, where a value of 1 indicates perfect agreement with expert annotation. A deep learning classifier, Loc-CAT, was trained on these labels. The resulting system achieved an F1 score of only 0.72 for subcellular localisation (Sullivan et al. 2018). This plateau highlights the limitations of crowdsourced labelling and points to a fully automated pipeline, preceded by a quality filter, as the more scalable solution.

2.2 Convolutional Networks And Foundation Models

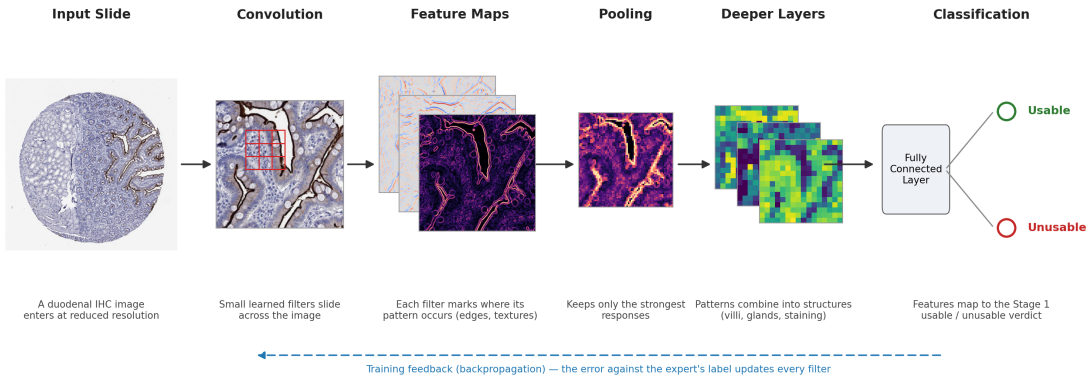

Figure 2-2: The stages of a convolutional network applied to one of the project's images, ending at the Stage 1 verdict. The feature maps are illustrative filter responses computed from the image shown, which is from the Human Protein Atlas (Uhlén et al. 2015).

Convolutional neural networks (CNNs) extract visual features from labelled images by passing them through multiple layers of learned filters, each capturing increasingly complex patterns (LeCun et al. 1998; Krizhevsky et al. 2012). This process is illustrated in Figure 2-2, which uses one of the project's small-intestine images as an example. A series of landmark achievements shaped the adoption of CNNs in pathology. The CAMELYON16 public benchmark competition tasked research teams with developing models that could detect lymph-node tumour deposits that have spread from the original cancer. The top-performing model matched the accuracy of an expert pathologist who examined the same images without time constraint, and outperformed a panel of eleven pathologists working under routine time pressure (Ehteshami Bejnordi et al. 2017).

The training of these models is shaped by a key practical constraint. Training a CNN from random initialisation requires a large, labelled dataset. For most pathology tasks, such datasets are rare because each one needs expert annotation of its own images. The accepted solution is transfer learning, in which a network trained on a broad image collection is repurposed as a starting point and then fine-tuned for the target task, eliminating the need to relearn general features. ResNet-50 is a common choice for this approach (He et al. 2016).

Residual connections allow ResNet-50 to be trained at a depth of fifty layers. This enables training without the accuracy decline that previously limited deeper models. Furthermore, its ImageNet-pretrained weights are widely available through standard libraries such as timm (Wightman 2019), making adoption in new research straightforward. As a result, ResNet-50 frequently serves as a reference architecture for transfer learning, providing a benchmark against which newer models are evaluated (Kornblith et al. 2019).

Despite this widespread adoption, ImageNet pretraining does not always provide a benefit. Raghu et al. (2019) demonstrated that transferring features from natural photographs offers limited improvement for two large-scale medical imaging tasks. They also found that much smaller models can perform comparably to large ImageNet architectures, and that some of the perceived improvement is attributable to over-parameterisation rather than truly reusable features.

A promising alternative is to pretrain networks directly on pathology images. CTransPath, pretrained on fifteen million histopathology patches, outperforms ImageNet-based features on various benchmarks, while UNI extends the same strategy to more than one hundred million images (Wang et al. 2022; Chen et al. 2024). PLIP takes a different direction by training on pathology images paired with descriptive text, allowing both images and language to be embedded in a shared representation, a design known as a vision-language model (Huang et al. 2023). However, their training datasets consist predominantly of H&E-stained images, so it remains uncertain whether their advantages carry over to tasks involving IHC-stained slides.

Another set of candidate models originates from object detection. The YOLO networks, coined for their principle of examining an image only once, are designed to balance speed and accuracy (Redmon et al. 2016). Recent versions include a compact classification variant (Jocher et al. 2023). In classification mode, YOLOv8 achieved 96.6% accuracy on gastrointestinal histology tiles (Farsi et al. 2025). However, a breast-cancer study reported variable performance, with the YOLO classifier outperforming ResNet-family baselines in some cases and falling short in others (Abd El-Hafeez et al. 2025).

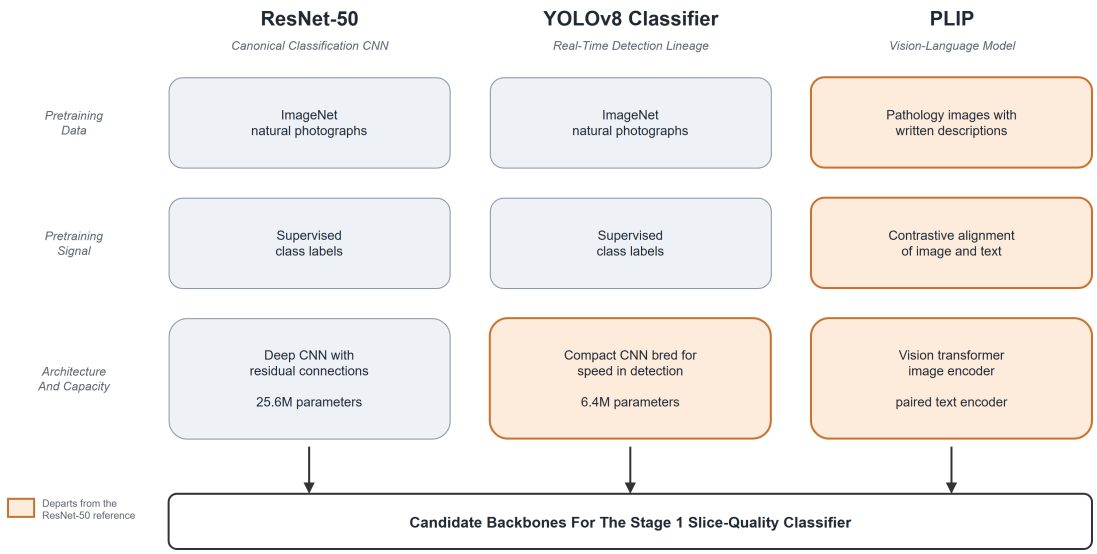

Figure 2-3: The three candidate backbones compared, with tinted cells marking departures from the ResNet-50 reference.

The three candidate models are fundamentally distinct, as summarised in Figure 2-3.

2.3 Slice-Quality Assessment And Artefact Detection

The performance of any image classifier is fundamentally constrained by input quality, and in histology, high-quality input is not always assured. Tissue sections are prone to artefacts introduced during preparation and scanning. Common artefacts include folds, tears, and uneven staining (Taqi et al. 2018). Schömig-Markiefka et al. (2021) demonstrated the impact of these artefacts by subjecting a trained pathology model to nine different types. Their experiment revealed that severe artefacts of any variety can substantially reduce model accuracy. Identifying images too damaged to analyse before any analysis begins is called quality control.

Automated quality control has evolved from fixed rules to data-driven models. HistoQC, for example, applies a sequence of image filters to flag artefacts and tissue folds in H&E sections (Janowczyk et al. 2019). These rules are transparent but manually set, so they do not adapt to new stains or scanners. More recent

systems use models trained on examples of poor quality rather than relying on hand-set rules. DeepFocus, a convolutional network that pinpoints out-of-focus regions on a slide, is one such example (Senaras et al. 2018). Its training set included both H&E and IHC slides, making it different from most quality tools, which are developed on H&E alone. However, DeepFocus only assesses focus, so it cannot flag folds, tears, or missing tissue.

The closest example of a whole-image quality assessment is PathProfiler (Haghighat et al. 2022). This multitask model predicts several slide properties at once, such as focus and staining quality. One output is a usability score that estimates whether a slide is suitable for diagnostic use. That score closely agrees with pathologists' judgement. Agreement was measured with the Pearson correlation, a coefficient that captures how consistently two sets of scores rise and fall together. Against the scores of three pathologists, PathProfiler's usability score reached a correlation of 0.89. This makes it the firmest precedent for the binary usable-or-unusable verdict a quality-control gate demands. Notwithstanding this, it was validated only on prostate tissue stained with H&E, leaving its reach across other stains and tissues untested.

Only one published study has used a YOLO network for slide quality checking. Hemmatirad et al. (2022) employed a YOLO detector to find artefacts on digitised slides, marking air bubbles and tissue folds. It answers a different question from a quality-control gate. Locating a fold indicates where a problem is, but does not determine whether the entire slide is suitable for analysis.

A recurring limitation emerges across these systems. Jurgas et al. (2024) note that quality-control methods perform well on H&E images, but can be less effective when applied to IHC. This is not surprising given that nearly all tools in this field were developed and validated using H&E imagery. Research focused specifically on intestinal tissue is even more limited. The closest example is from Schreiber et al. (2024), who removed pen marks and scanning artefacts from small-intestine biopsy images and segmented the remaining tissue. Their approach, even so, rests on classical image processing rather than deep learning, and was likewise devised for H&E. To the author's knowledge, no published system, whether rule-based or model-based, currently provides a usability assessment for IHC images of intestinal tissue.

2.4 Learning From Scarce And Imperfect Labels

Training data influences a medical classifier as much as its architecture does. For this type of project, the available labels are often scarce, imbalanced, and grouped by donor. Imbalance matters most when the rare class is the focus. A model that consistently predicts the majority class may achieve high overall accuracy, yet still fail to detect the images that a quality gate is intended to identify. To counteract this, effective strategies include oversampling the minority class, adjusting the decision threshold, and reporting per-class results rather than relying on a single accuracy metric (Buda et al. 2018). Grouping is also crucial because images from the same person should never appear in both the training and testing datasets. If a model sees the same donor in both sets, it may learn to recognise the donor rather than the relevant property being classified. Tampu et al. (2022) demonstrated in medical imaging that this can inflate reported accuracy by as much as thirty per cent.

Two main approaches address label scarcity. One method is to focus expert effort strategically, using the current model to select the most informative examples for annotation rather than labelling at random (Budd et al. 2021). The second approach is to reduce the need for human labelling altogether. Recent advances show that large language models can outperform crowd workers on text annotation tasks (Gilardi et al. 2023), and vision-language models such as PLIP connect pathology images to descriptive text (Huang et al. 2023). Nonetheless, researchers have chiefly tested large language models on text, and PLIP learned from social-media captions rather than expert annotation. More labels do not necessarily mean better quality, and it remains unclear how much machine-generated labels can replace expert annotation.

2.5 Region Localisation From Slide-Level Labels

Stage 2 addresses the challenge of identifying analysable tissue within a usable image, even when no region-level annotations are available. The literature offers a solution through multiple instance learning (MIL), where each image is divided into tiles, and only the group (or 'bag') is labelled, not the individual regions. Campanella et al. (2019) demonstrated that classifiers can be trained accurately

using only slide-level labels at clinical scale. Ilse et al. (2018) introduced attention into MIL, enabling the model to learn the importance of each tile for the overall prediction. Lu et al. (2021) further showed that these attention weights can be visualised as heatmaps that highlight the diagnostic tissue. Localisation thus surfaces as a by-product of classification; all the same, the heatmap captures what the model deems discriminative, not what an expert would delineate, and without region annotations, the level of agreement between the two cannot be assessed.

Selecting a localisation method depends on the specific requirements of each approach. Segmentation by fixed rules works when the target is clearly distinguishable from its surroundings. For example, Schreiber et al. (2024) removed pen marks and scanning artefacts from small-intestine images using one such rule. Their method transforms the image colours so tissue separates from artefacts with a single threshold, which works because a pen mark has a colour not found in normal tissue. For all that, the distinction Stage 2 must draw is subtler. No such colour difference separates readable epithelium from unreadable epithelium; both are the same tissue under the same stain. What distinguishes them is how the section was cut and how intact its structure remains. This type of judgement is best made by an expert rather than a fixed rule. A detection model such as that of Hemmatirad et al. (2022) offers an alternative, inferring the judgement from annotated examples. Nevertheless, this method demands region-level annotation for every training image, reviving the expert-labelling burden the pipeline seeks to avoid. By contrast, multiple instance learning (MIL) needs only image-level labels, like those a quality classifier already uses.

Previous research on localisation in HPA imagery has focused on different aspects of the problem. Two public challenges required models to classify subcellular patterns in immunofluorescence images of cultured cells (Ouyang et al. 2019; Le et al. 2022). Even the winning models scored modestly; the first challenge's winner reached a macro-F1, the average of per-class F1 scores, of only 0.59. DeepHistoClass advanced the field by moving the task to tissue IHC, classifying 7,848 testis images (Ghoshal et al. 2021). This demonstrated that such models can operate on tissue as well as cultured cells. To the author's knowledge, the only study to address apical-basal polarity directly classified cultured spheroids instead of tissue, achieving agreement with trained human raters in 86.3% of cases

(Soetje et al. 2020). Accurately reading a protein's compartment in intestinal tissue depends on first identifying the readable epithelium. Be that as it may, to the author's knowledge no published system yet furnishes this localisation.

2.6 Synthesis

This review reveals a clear gap in the current literature. The Human Protein Atlas (HPA) has outgrown manual annotation, and even large-scale crowdsourcing has reached its limits (Sullivan et al. 2018). While computational analysis tools are mature, nearly all have been developed for H&E-stained images. Quality-control systems are limited to H&E, artefact detection, or fall short of providing usability assessments for IHC images. Similarly, localisation methods have been applied to cultured cells and other tissues but not to the intestinal epithelium. No published system identified in this review, across any architecture, classifies slice quality on intestinal IHC or localises analysable tissue within such images using only slide-level labels. The planned two-stage pipeline extends the usability-verdict precedent to this staining and tissue (Haghighat et al. 2022). Whether established methods survive this transfer is the question the project exists to answer.

Chapter 3

Data Engineering And Curation

3.1 The Pipeline And Its Guiding Principles

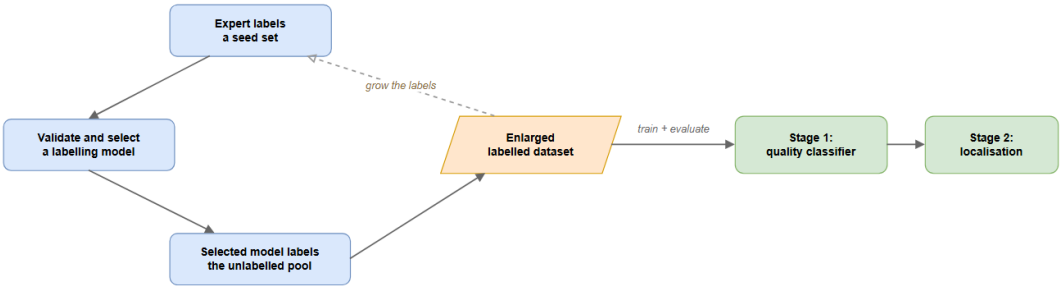

Figure 3-1: The data and modelling pipeline. Expert labels select a labelling model, which expands them into a larger set that trains the Stage-1 classifier and feeds the Stage-2 localisation.

The dataset was constructed in sequential stages, each anchored to expert-assigned labels. This chapter traces the entire data pipeline, from raw image acquisition through dataset preparation for model training and evaluation, as shown in Figure 3-1.

The first guiding principle is that the pipeline is non-destructive. The pipeline simply labels unusable slides; it never moves, alters, or deletes original images. Each step reads the table from the previous stage and writes a new one, rather than modifying the data in place. Source images are never modified, so any step can be rerun or revised without repeating the ones before it.

The second principle is reproducibility. The image filename encodes its own metadata and is the join key for every table, so the dataset can be rebuilt from the raw export alone. The randomised steps in training are initialised from a fixed value called a seed, so they repeat identically on every run. Acquisition relies on a single local copy of the HPA export rather than a live external query, so the same image set can be reproduced from scratch.

The third principle involves the cost of error, which is central to both labelling and evaluation. Passing an unusable section introduces corrupted data that can affect downstream analysis. Rejecting a usable section only results in the loss of that image. Since corrupted data is the graver risk, the pipeline is tuned to reject when uncertain.

3.2 The Source Of Tissue Samples

Each image in the small-intestine set represents a single tissue core, defined as a small circular section of tissue extracted from a donor and stained for one antibody target (Uhlén et al. 2015). The scanned resolution is approximately 3000 by 3000 pixels, corresponding to the atlas's 20× scanning magnification (Kampf et al. 2012). As each core is stained for a single antibody, every image maps to one gene. The downloaded pool contains approximately 3,100 images, with most originating from the duodenum. Each gene has about two cores on average, yielding coverage of roughly 1,550 genes. This dataset is broad in gene representation but shallow in samples per gene.

The atlas does not record slice quality. Its annotation describes each gene rather than the readability of individual sections (The Human Protein Atlas 2022), so no field indicates whether a core is suitable for analysis. Therefore, the required label was constructed specifically for this project.

3.3 A Self-Contained Downloader

Table 3.1: The metadata fields read from each Human Protein Atlas XML entry. The first five form the image filename.

Field	XML element	Role
Tissue	`tissueDescription`	selects small-intestine entries
Gene	`name`	the antibody target
Sex	`sex`	donor sex
Age	`age`	donor age
Donor ID	`patientId`	the donor-grouped evaluation key
Image	`imageUrl`	location of the core image fetched

Data acquisition is managed by a custom downloader developed for this project. This tool reads a single local copy of the atlas's small-intestine XML export, downloaded in July 2026 (The Human Protein Atlas 2026a), and processes it entry by entry. For each entry, the downloader extracts the recorded metadata, summarised in Table 3.1. To improve efficiency, only small-intestine samples are retained, and the number of images per gene is capped. Each matching image is fetched and saved under a filename constructed from its metadata, ensuring that each file records its donor of origin.

The implementation is intentional. Acquisition is performed as a single, self-contained step: one script reads from one local copy of the export. Retaining this copy guarantees the same image set can always be reproduced. The donor identifier is included in every filename, providing each image with a traceable origin.

3.4 Naming Scheme And Metadata Joins

Every image is saved under a filename built to a fixed scheme which includes the tissue, gene, sex, age and donor identifier.

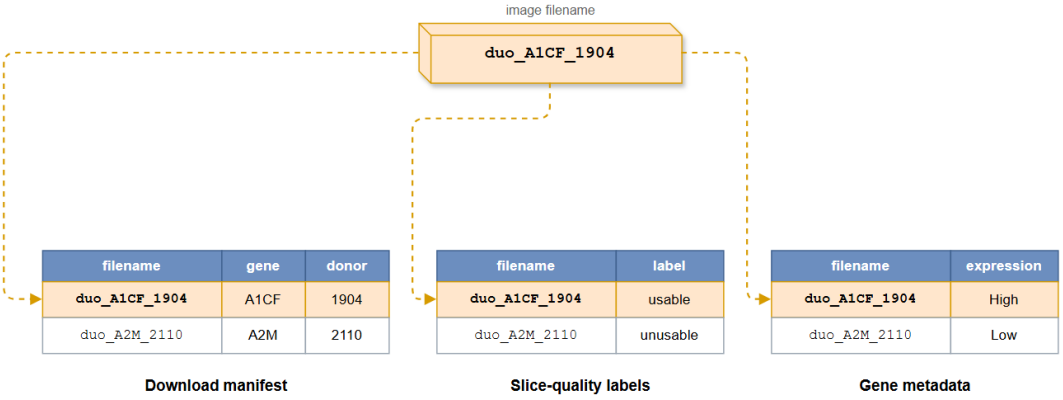

Figure 3-2: The same filename keys this image's row in every table, so the tables join on that name alone.

The filename ties the pipeline's tables together, as Figure 3-2 shows for a single image. Each stage writes its own table, in which every row carries the image's filename. The download table records gene and donor, while the labelling table adds the verdict. The third table holds gene-level attributes such as expression level and antibody reliability. Because the tables share this key, one image's rows can be matched across all of them without a separate index to maintain.

3.5 Data Audit

Training a model on faulty data introduces persistent challenges for all subsequent results. To mitigate this risk, a thorough data audit was conducted prior to classifier training. The export utilised was the atlas's normal-expression dataset, so every sample was initially presumed to be normal. In point of fact, the export does record each sample's tissue state (normal or diseased, for instance). For this project, the downloader disregarded morphology and other unexamined fields, such as expression levels and antibody reliability, as none were used in the selection process. The downloader focused exclusively on small-intestine terms, such as duodenum. If detected, the image was downloaded. This approach resulted in two main issues.

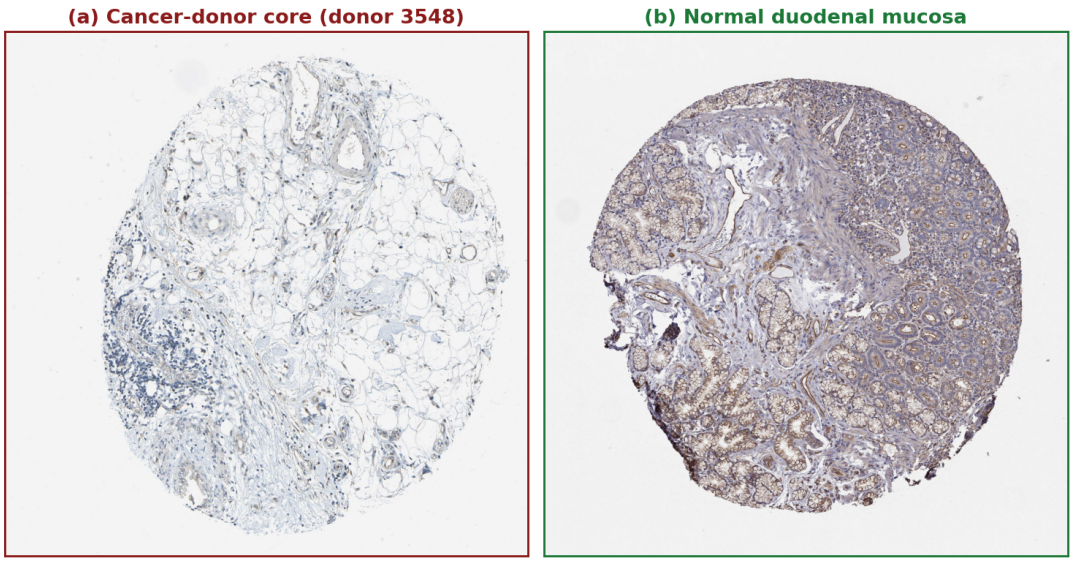

Figure 3-3: A core from cancer-carrying donor 3548 (left) beside normal duodenal mucosa (right). Selection by topography alone allowed such cores into the collection. Both cores are Human Protein Atlas images (Uhlén et al. 2015).

The first issue was contamination arising from these assumptions. Each sample in the export includes a set of SNOMED annotations. SNOMED, Systematized Nomenclature of Medicine, is a standard clinical vocabulary, assigning fixed codes to every medical concept (Cornet and Keizer 2008). Such as anatomical sites or diseases.

Table 3.2: The sample morphology codes carried by the `snomedCode` attribute, which the downloader never read. Codes in the neoplasm range (M-8xxxx and M-9xxxx) mark contamination in a normal-tissue set.

Morphology code	Description in the export	Status in a normal-tissue set
M-00100	Normal tissue, NOS	expected
M-81403	Adenocarcinoma, NOS	contamination
M-82403	Carcinoid, malignant, NOS	contamination

Some annotations specify the tissue's topography, which is the anatomical site of origin. While others describe morphology. By not checking each sample's condition, the downloader inadvertently included tissue from donors with cancer. The difference between tissues is shown in Figure 3-3. Varying labels include

'adenocarcinoma' or 'carcinoid', as summarised in Table 3.2. This contamination matters because the dataset is intended to represent normal small intestine tissue. Consequently, a model trained on it would learn features partly from cancerous samples, undermining its validity.

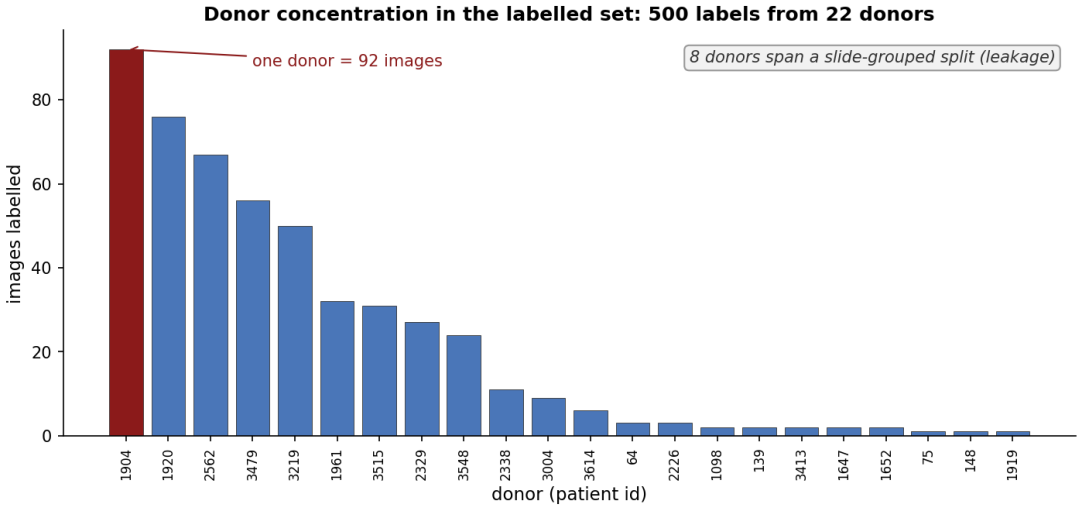

Figure 3-4: The number of images each donor contributed to the 500 labels. Donor 1904 alone accounts for 92 of them.

The second issue concerned the narrow donor base from which the labels were derived. Of the 500 labelled images, only 22 unique donors were represented. A single donor identified as 1904 supplied 92 images alone, as shown in Figure 3-4. This imbalance arose because the downloader limited the number of images per gene, but not per individual. With the atlas's small-intestine collection drawn from just 59 donors, many genes rely on tissue from the same individuals. Notably, donor 1904, who contributed the most images, was also among the cancer-carrying donors. Compounding the contamination and donor-imbalance issues.

This has significant implications for model evaluation. For a fair assessment, a classifier should be tested on data from donors it has not seen during training. In the event, the dataset was split image by image rather than by donor. As a result, eight donors had images in both sets, enabling the model to potentially recognise tissue from familiar donors instead of genuinely assessing slice quality. This phenomenon, known as data leakage, artificially inflates model performance (Kaufman et al. 2012). Importantly, this issue cannot be resolved simply by collecting more data, as only 59 small-intestine donors exist in the atlas, of whom

just 43 are cancer-free. All subsequent data divisions in this project were therefore constrained by these intrinsic limitations.

3.6 How The Audit Changed Acquisition

The audit's findings prompted two key corrective actions. First, all donors whose atlas records indicated a neoplasm morphology code were excluded from the dataset. A neoplasm, more commonly known as a tumour, is an abnormal growth of tissue, so no donor bearing such a code could contribute to a normal-tissue set. Sixteen such donors were identified in the export, and all their images were removed from consideration. Consistent with the pipeline's non-destructive principle, this exclusion was recorded as metadata rather than through physical deletion, ensuring the affected images remain on disk and the decision can be revisited if necessary. This measure eliminated contamination, but further reduced an already limited donor pool.

The second action involved a revised download protocol to prevent recurrence of the identified issues. The updated downloader capped acquisition at twelve images per donor, rather than two per gene, thereby distributing the collection more evenly across individuals. Additionally, tissue condition was now checked prior to downloading; any sample with a neoplasm code was excluded, and the sixteen previously identified donors were omitted entirely. Donors already included in the dataset were also skipped, ensuring each new image originated from a unique individual. This process resulted in the addition of 224 images from 19 cancer-free donors.

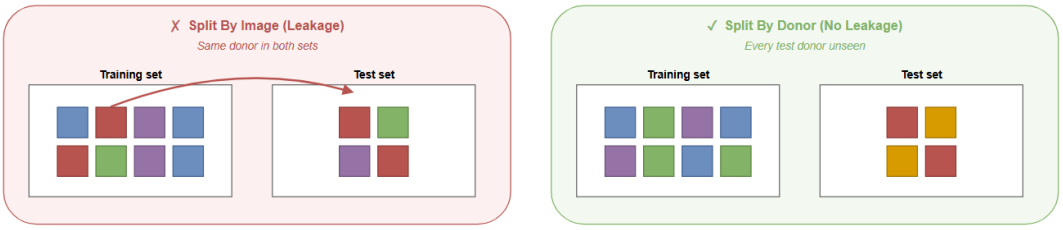

Figure 3-5: Each square is one image and each colour is one donor. Splitting by image (left) lets the same donor's images reach both the training and test sets, so the model meets familiar donors at test time and its score is inflated. Splitting by donor (right) keeps each donor wholly on one side, so every test donor is genuinely unseen.

Together, these actions produced a corrected dataset encompassing all 43 cancer-free donors available in the atlas. While the imbalance from earlier collections could not be retroactively addressed, the new additions are distributed evenly, with no more than twelve images per donor. Each image continues to carry its donor identifier in the filename, enabling future data splits to be performed on a per-donor basis and directly mitigating the leakage risk identified by the audit. Figure 3-5 contrasts the two approaches; an image-level split allows one donor's images to appear on both sides of the division, whereas a donor-level split keeps every test donor entirely unseen. Although the maximum of 59 donors remains, within this constraint the collection is now as broad and as representative as the atlas permits.

Chapter 4

The Labelling Decision And Annotation Tool

Before any labelling could begin, it was essential to establish a precise, objective criterion for annotation. This led to the development of a rubric. A rubric is a fixed set of written rules specifying exactly how each judgement should be made, ensuring that every slide is assessed according to the same standard. Because the atlas does not provide a quality label, the rubric prevents the project's labelling from relying on subjective impressions, which can drift over time and introduce noise into the training data. Moreover, the rubric enhances transferability. Any assessor, whether human or automated, can apply the same criterion. The rubric itself was developed collaboratively with the project's domain expert.

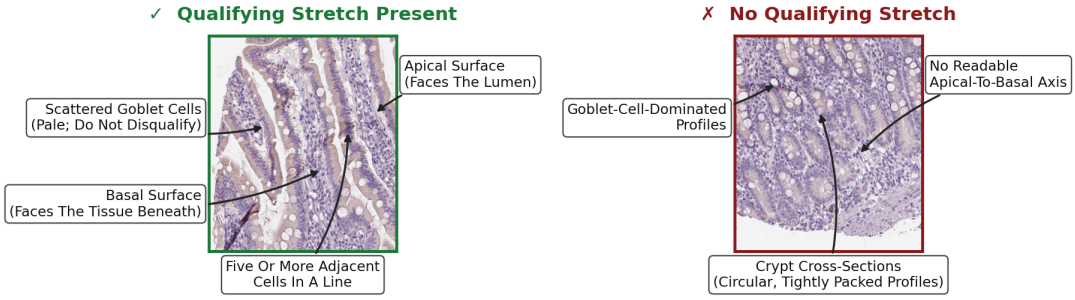

Figure 4-1: A qualifying stretch (green) beside a region without one (red). The image is from the Human Protein Atlas (Uhlén et al. 2015); the annotations were added for this project.

Under this rubric, each slide is evaluated as a complete unit at native res-

olution and assigned one of two labels: usable or unusable. The criteria are intentionally straightforward. A slide is considered usable if it contains at least one qualifying stretch of epithelium anywhere on the slide, regardless of orientation, as shown in Figure 4-1. If no such stretch is present, the slide is deemed unusable. The state of the remaining tissue is irrelevant, as poor or complex regions elsewhere do not influence the judgement.

A qualifying stretch must fulfil three conditions. First, it should include at least five adjacent epithelial cells aligned in a clear linear arrangement. Second, both the apical and basal surfaces must be visible. Third, staining must be consistently readable along the entire stretch. The stain colour is not important. Either the brown antibody signal or the blue nuclear counterstain is sufficient. The five-cell threshold supports repeatability in human judgement. Importantly, the classifier is intended to learn the visual pattern of a qualifying stretch, not to count cells explicitly. The classifier is never given the rule itself; it sees only images paired with the expert's verdicts and learns whichever visual features distinguish the two.

Notably, this definition frames unusability as a feature of epithelial architecture, rather than as a consequence of image artefacts such as folding or blurring. Most slides fail not because of technical flaws, but because the epithelium cannot be read along the apical-to-basal axis. For example, Figure 4-1 depicts a tissue sample that, at first glance, might appear unusable. However, the left region contains a qualifying stretch, while the right region contains none. A common failure arises from crypt cross-sections. A crypt is a small tube of gland cells at the base of the intestinal lining. When a section passes horizontally through a crypt, the cells form tightly packed rings. In these cases, no five cells align in one direction, so a qualifying stretch cannot form.

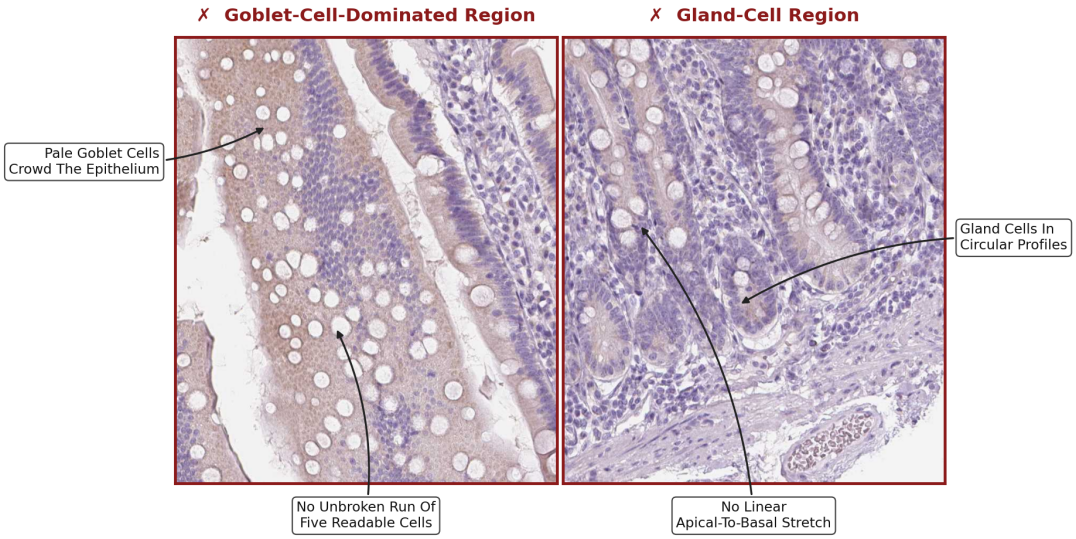

Figure 4-2: Two failing regions, dominated by goblet cells (left) and gland cells (right). The image is from the Human Protein Atlas (Uhlén et al. 2015); the annotations were added for this project.

Another common failure occurs in regions dominated by gland or goblet cells, as shown in Figure 4-2. Goblet cells are mucus-filled epithelial cells that appear pale in the stain. While scattered goblet cells are present in healthy stretches and do not disqualify them, their predominance in a region causes inconsistent staining and prevents an uninterrupted run of five readable cells. Another failure mode is the transition zone at the crypt mouth, where a mixture of cells at different developmental stages makes reliable reading impossible. In ambiguous cases, the default is to label the slide unusable. Consequently, the rubric effectively reduces each judgement to a single guiding question, promoting consistent decisions across hundreds of slides.

4.1 The Expert Annotation Platform

Implementing the rubric across hundreds of images required a dedicated system for recording expert judgements. To meet this need, the system had to satisfy three requirements. First, each image needed to be accessible as a complete unit at its native resolution. Second, each judgement was required to be a single binary decision. Third, the expert needed to review images remotely without installing

specialised software. While established open-source platforms such as QuPath offer whole-slide image analysis with region-level annotation tools (Bankhead et al. 2017), their generality exceeds the specific needs of this task, which requires exactly one verdict per image and no setup cost for the reviewer. To satisfy these requirements, a lightweight web application was developed specifically for the labelling campaign.

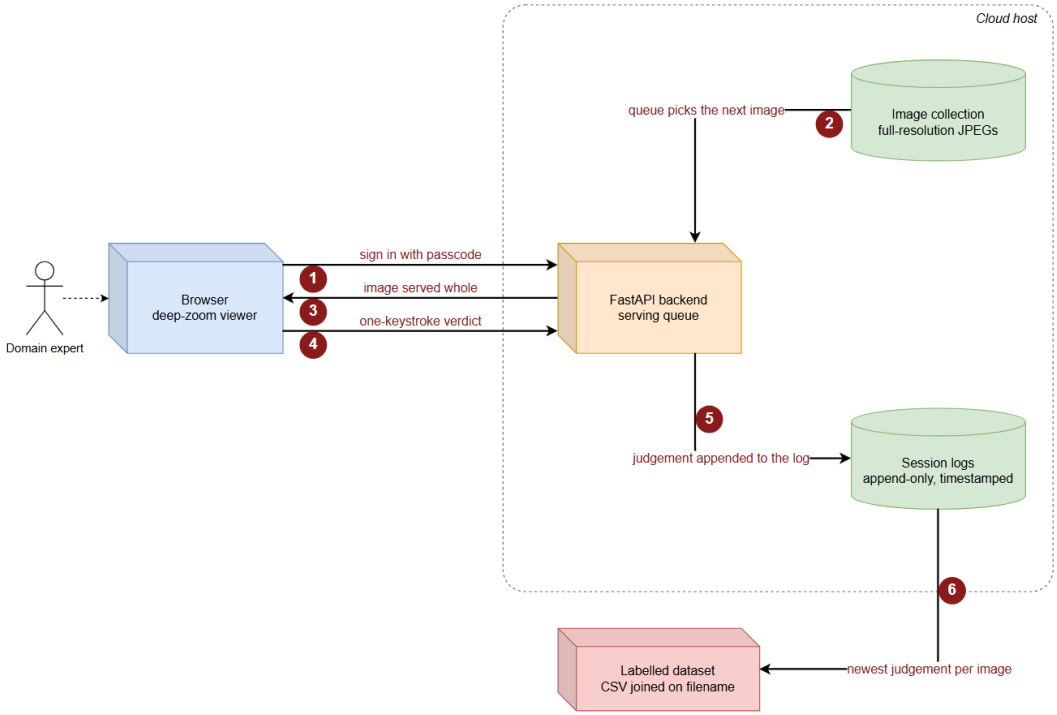

Figure 4-3: The annotation platform's client-server architecture. The numbered arrows follow one judgement from sign-in to the exported dataset; steps two to five repeat for every image.

The platform utilises a client-server architecture, as shown in Figure 4-3. The server component comprises a single backend service implemented using FastAPI, a Python framework for building web services (Ramírez 2018). This service performs three primary functions. First, it stores the image collection and delivers each image in its entirety at the scanned resolution. Second, it manages the serving queue, which tracks images that the expert has not yet reviewed. Third, it appends every incoming decision to a timestamped session log, writing to disk as each verdict is received. No entries in this log are overwritten; an undo operation

appends a reversal row rather than deleting an existing entry. When records are accessed, the most recent judgement for each image is taken as the current label. Completed judgements are exported as a comma-separated file keyed on the image filename, joining every label to the donor and tissue metadata carried in each filename. The platform's user and maintenance guide is provided in Appendix A.

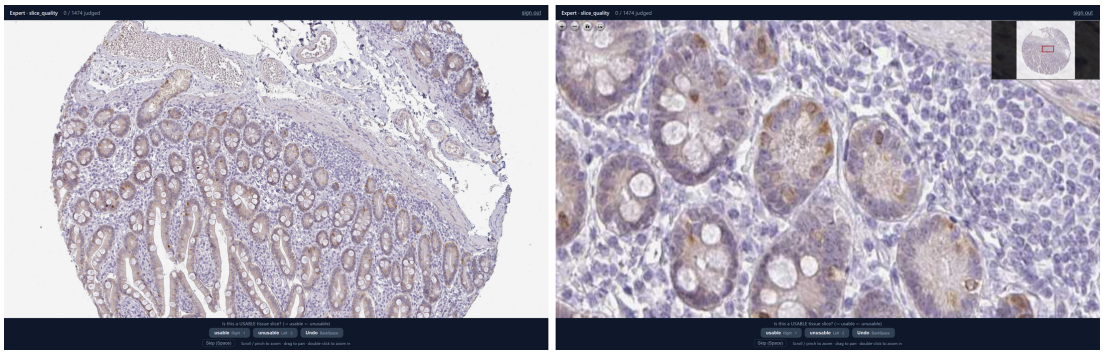

Figure 4-4: The platform's judging view, showing a full tissue section awaiting its verdict beside the same section under deep zoom, with the navigator marking the inspected region. The tissue image is from the Human Protein Atlas (Uhlén et al. 2015).

On the client side, the browser hosts a single-page interface that displays one image at a time, as illustrated in Figure 4-4. Each image is rendered using the OpenSeadragon deep-zoom viewer (OpenSeadragon contributors 2025). This viewer enables the reviewer to pan across the section and smoothly magnify any region, guaranteeing that the rubric's requirement to judge the image as a complete unit is met with full detail.

To begin, the expert requires only a browser, a name, and a passcode, with no software installation or file management necessary, as the platform is hosted in the cloud. Minimising the time and effort required for each judgement was the platform's primary design objective. Pressing a single key records the verdict and automatically loads the next image for review. If an accidental error occurs, the reviewer can undo the action. Each judgement is written to permanent storage on the server immediately upon submission, making sure that no data is lost if the browser is closed. Upon return, the queue restores the reviewer's position.

4.2 Assessing Label Reliability With A Single Annotator

The domain expert judged each image alone, without a second rater for comparison. This design carries a known limitation. A common safeguard in labelling studies is to have two experts label the same images independently and measure their agreement, which indicates whether the labels reflect more than one person's opinion. With a single expert, this check is unavailable. The only measurable property is whether the expert, shown the same image again, reaches the same verdict. Cohen's kappa is a standard instrument for this measurement. It is an agreement statistic that corrects the raw proportion of matching verdicts for the agreement two raters would reach by chance (Cohen 1960). The formula for calculating kappa is as follows:

$$\kappa = \frac{p_o - p_e}{1 - p_e} \tag{4.1}$$

Where:
p_o is the observed proportion of matching verdicts
p_e is the agreement expected by chance

For this project, label reliability depends on consistency. The rubric is designed to remove personal opinion from the decision process, making it vital to verify that the expert applies the rule uniformly in every case. To assess this, the domain expert re-evaluated 50 slides from the completed set without knowing they were repeats, ensuring independent second judgements. The expert's two assessments matched on 47 out of 50 slides, yielding a raw agreement of 94 per cent and a Cohen's kappa of 0.735. According to conventional benchmarks, a kappa between 0.61 and 0.80 indicates substantial agreement (Landis and Koch 1977). Given that the expert replicated their own verdicts on 94 per cent of slides, it is improbable that another labeller, whether human or automated, would achieve closer agreement with these labels than the expert does with themselves.

4.3 Scaling The Labels With A Vision-Language Model

After the annotator validated the dataset using the annotation tool, it became clear that there were not enough unusable images to train an effective classifier. The gold dataset consists of 847 images, but only 163 are labelled as unusable, as summarised in Table 4.1. This minority class presents a challenge, as the classifier must learn the features of unusability from these limited examples. Since it was not possible to increase expert labelling capacity, the solution was to expand the image pool. A total of 5,900 images were downloaded from the atlas's small-intestine collection (The Human Protein Atlas 2026a), excluding those from cancer-carrying donors. Machine annotation was then used to address this bottleneck (Gilardi et al. 2023). This approach worked because the rubric served as a prompt, the written instruction a model receives. The labeller could be validated against a set of expert-labelled images before its output was accepted into the larger pool. This kept the gold dataset as the quality benchmark throughout the expansion.

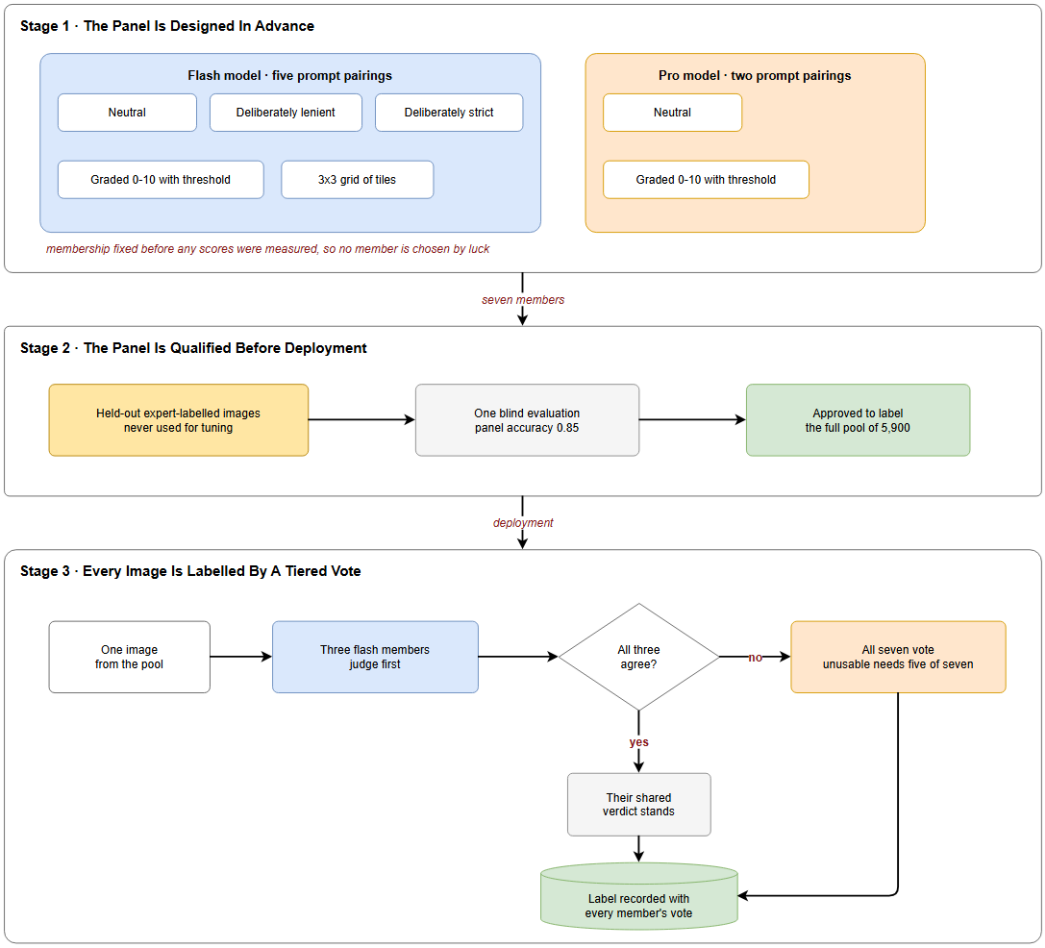

Figure 4-5: The three stages of the machine-labelling design. The panel is fixed in advance, qualified once against held-out expert labels, and then labels every image by a tiered vote.

The labelling pipeline is built on Gemini 2.5, a vision-language model available in a faster Flash and a more capable Pro variant (Comanici et al. 2025). The model receives the rubric verbatim and assesses each image in full, the same task the domain expert performed. A single model reading a single prompt, however, errs systematically rather than randomly. Every single-prompt configuration tested in development carried a consistent bias, and the strictest flagged roughly a third of truly usable images as unusable. To counter this, each image goes before a panel of seven judges whose biases differ by design, as illustrated in Figure 4-5, so that their errors cancel in the vote. Five run on Flash with distinct

prompt styles, including one that reads the image as a grid of tiles. The remaining two run on Pro. Three Flash judges assess each image first. A unanimous verdict stands. Disagreement marks the image as borderline and refers it to the full panel, whose vote is final. The full panel therefore runs only where the first stage cannot decide.

The panel's membership was decided before any scores were consulted. The alternative, keeping whichever combination scored highest on the expert-labelled evaluation images, would have rewarded luck; with only a few hundred images, the best-scoring panel is tuned to their quirks and its figure does not repeat on fresh data. An earlier selection round demonstrated this, as an apparent accuracy of 0.91 fell to 0.81 on new data. Fixing the membership in advance keeps the panel's measurements honest.

Before being applied to the full dataset, the panel was validated once on a held-out split of the expert-labelled images, achieving 85 per cent accuracy. Only after this assessment did it proceed to label all 5,900 images. The process was completed within hours and without errors. Each label includes the underlying vote distribution, allowing unanimous and close decisions to be distinguished.

4.4 The Gold And Silver Datasets

Table 4.1: The labelled datasets produced by the annotation campaign and their roles.

Dataset	Produced by	Images	Usable/ unusable	Donors	Role
Gold	The domain expert, through the annotation platform	847	684 / 163	39	Trains models; the sole evaluation ground truth
Silver	The rubric-driven model panel	5,900	5,178 / 722	43	Additional training volume only
Confident subset	Silver images with near-unanimous votes	4,249	3,952 / 297	—	The silver slice trusted for training

This approach resulted in two labelled datasets with distinct origins and status, as summarised in Table 4.1. By convention, expert-verified labels constitute the gold dataset, while machine-generated labels form the silver dataset. Within the silver dataset, the 4,249 images with near-unanimous panel votes are considered the confident subset, the most trusted tier of the machine labels.

Where the datasets overlap, the expert's label takes precedence. In total, 759 images have both expert and machine labels. The datasets' asymmetry extends to their roles. The silver dataset is used solely to increase training volume, while all evaluations rely exclusively on gold. Therefore, the expansion increases the range of data available for learning without altering the standard used for assessment.

Chapter 5

Slice-Quality Classifier

Two candidate architectures were trained on all the training datasets that were made for this project. In addition, they were compared using a single, fixed evaluation protocol. The protocol is described first, as all subsequent results rely on its specifications.

5.1 Evaluation Protocol

All models in this chapter are evaluated using a single, fixed protocol to allow direct comparison of results. Evaluation is performed with five-fold cross-validation. The labelled images are split into five groups, or folds. Each fold serves as the test set once, with the other four as training data. Folds are grouped by donor, ensuring no donor's images appear in more than one fold. The five test folds are drawn from the 847-image gold dataset. Moreover, some later models use the machine-labelled images as additional training data; however, these images never enter a test fold.

Importantly, every model in the final comparison uses the identical fold partition. Some folds are naturally harder than others, because the donors they contain differ in how ambiguous their tissue is. In this dataset, the resulting fold-to-fold swing is comparable in size to the differences between the models themselves. If each model received its own random split, a higher score could therefore mean nothing more than an easier set of test donors. Sharing one partition removes this ambiguity, as each model faces the same test images in every fold. The random seed is fixed at 42, ensuring that the partition can be

regenerated exactly in any subsequent run.

Within each training fold, a portion of the data is set aside as a validation slice. This slice is also grouped by donor, so no donor appears in both the training portion and the validation slice. Every tuning decision is made on this slice. This includes when to stop training and where to place the decision threshold. The test fold itself is therefore touched once, when the final score is computed. Thus, aggregate results are reported as the mean and standard deviation over the five folds.

Choosing the metrics required care, because the dataset is imbalanced. About 81 per cent of the gold labels are usable, so a model that calls every image usable would achieve 80.8 per cent accuracy while being ineffective as a filter. For this reason, accuracy is reported but never used to judge a model. The primary metric is instead the macro-averaged F1 score, hereafter macro-F1. It is computed as

$$\text{macro-F1} = \frac{F1_{\text{usable}} + F1_{\text{unusable}}}{2}, \qquad F1_c = \frac{2\,P_c\,R_c}{P_c + R_c} \tag{5.1}$$

Where:
P_c is the precision of class c
R_c is the recall of class c

Averaging the two classes with equal weight ensures that the minority class counts as much as the majority (Buda et al. 2018).

Unusable recall, the term R_{unusable} in Equation 5.1, is additionally reported on its own. An average can conceal the one failure that matters here, as a model may hold a reasonable macro-F1 while missing many unusable images. Catching those images is the filter's purpose, so this component is tracked separately.

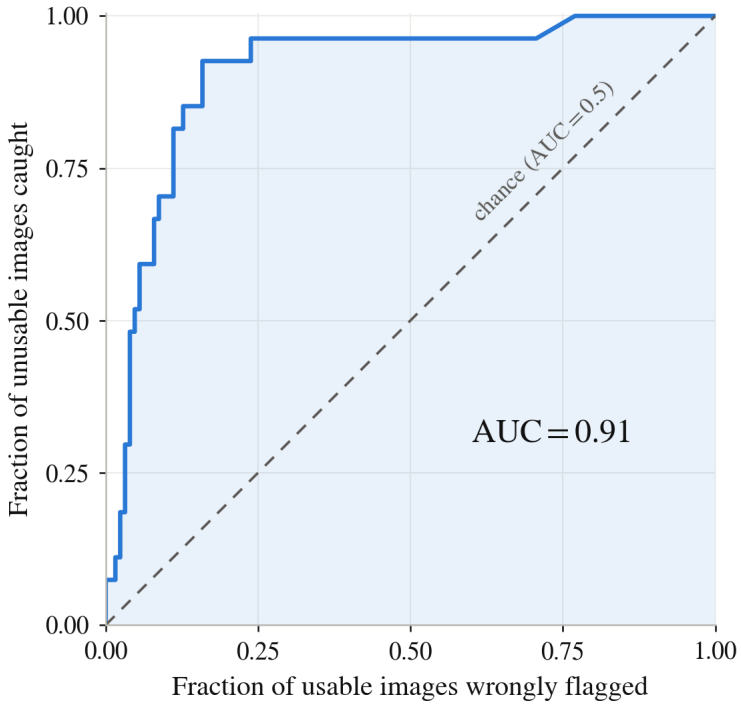

Figure 5-1: The receiver operating characteristic curve of the combined-data ResNet, computed from its per-image predicted probabilities on one test fold. The fold shown is the one whose AUC, 0.91, lies closest to the model's five-fold mean, and the dashed diagonal marks chance.

The area under the receiver operating characteristic curve (AUC) is also reported. AUC represents the probability that a randomly selected unusable image will be ranked as more likely unusable than a randomly selected usable image, and corresponds to the shaded area in Figure 5-1. This measures the model's ability to separate classes independently of any threshold.

Table 5.1: The two reference baselines, scored under the identical donor-grouped protocol as every model in the chapter.

Baseline	Decision rule	Macro-F1
The usable predictor	Labels every image usable	0.4462 ± 0.0149
Tissue-fraction heuristic	Labels an image usable when its tissue coverage exceeds a threshold chosen on the training folds	0.5656 ± 0.0508

Reported scores only carry meaning against a point of comparison. To provide one, two simple baselines are scored under the same protocol, as summarised in Table 5.1. The first is the usable predictor, which labels every image usable because usable is the more common class. Its 80.8 per cent accuracy collapses to a macro-F1 of 0.4462, since it catches no unusable images. The second is a heuristic that measures how much of each image is covered by tissue and labels the image usable when that coverage is high enough. It scores a macro-F1 of 0.5656. Any trained model must clearly exceed both before its added complexity is justified.

Converting a model's output into a verdict involves one additional choice. The network does not produce a verdict directly, but a probability that the image is unusable. A rule is therefore needed to turn that probability into one. This rule is the operating point. Every model is reported at two points. The first is argmax, which selects the more probable class. The second compares the probability with a threshold chosen on the validation slice to maximise macro-F1, never on test data. Scores under this rule are reported as tuned.

The two operating points do not give equal scores. Tuning the threshold on the validation set usually raises a model's score, because the threshold is chosen to suit that model's own probabilities. Comparing one model's tuned score with another's argmax score would measure the benefit of tuning rather than the difference between the models. Tuned scores are therefore compared only with tuned scores, and argmax scores with argmax scores.

5.2 Backbone And Architecture Selection

Three candidate backbones were carried forward from the earlier comparison. Each makes a different case for selection. ResNet-50 is the safest starting point, as its fine-tuning behaviour on small datasets is well understood (He et al. 2016; Kornblith et al. 2019). Its weakness is equally clear, since it learned its transferred features from everyday photographs rather than tissue (Raghu et al. 2019). PLIP makes the opposite case. Its features already encode the appearance of stained tissue (Huang et al. 2023). This means it needs less adaptation, provided the advantage survives the constraints under which it must be deployed. YOLOv8s-cls makes its case through its training pipeline rather than its weights (Jocher

et al. 2023). The pipeline includes augmentation, weight averaging and class balancing. It therefore tests whether disciplined training matters as much as the choice of network. These claims could not be settled by argument, so each was put to measurement.

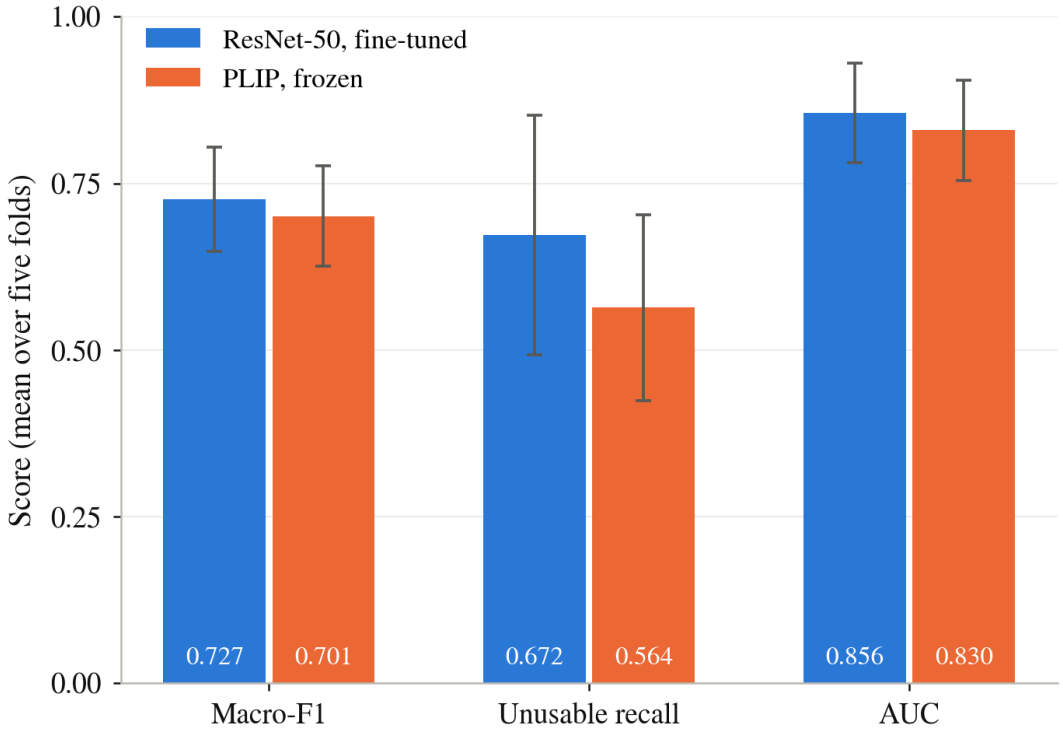

Figure 5-2: A bar chart of the frozen PLIP configuration against the fine-tuned ResNet-50 on identical folds, with error bars showing the spread across the five folds. PLIP trails on every metric, with a macro-F1 of 0.701 against the ResNet's 0.727.

PLIP was first evaluated in its frozen configuration, where the backbone weights remained unchanged, and only a simple classifier was trained on the extracted features. Using the protocol described above, it achieved a macro-F1 of 0.701 ± 0.075 and an unusable recall of 0.564 on the 847 gold-labelled images, as shown in Figure 5-2. In comparison, a fine-tuned ResNet-50 outperformed PLIP on all metrics across the same folds, with unusable recall 0.108 higher. This result excludes PLIP as implemented here rather than domain pretraining in general. The comparison set a frozen backbone at a smaller input size against a fully fine-tuned model. Because CTransPath and UNI are built on pathology pretraining

like PLIP (Wang et al. 2022; Chen et al. 2024), the negative outcome reduced the incentive for additional pretraining experiments. Regardless, their pretraining is dominated by H&E staining, which mismatches IHC as noted earlier. With the most accessible member of this family failing to beat the ImageNet transfer, the less suitable members were not evaluated.

YOLOv8s-cls remained in contention on different grounds. Its shipped pipeline trains with aggressive built-in augmentation, cropping a different region of each image at a different scale on every pass, and keeps a running average of the network weights (Jocher et al. 2023), offering a deliberate contrast to the controlled ResNet recipe. The pipeline was used exactly as shipped, meaning nothing in its configuration was altered. Moreover, a preliminary screening reached a macro-F1 of 0.8626. At first, it was a clear contender against ResNet-50; however, the experiment used an earlier partition that did not enforce donor grouping. To discover YOLOv8's true potential, further experimentation was needed. As a result, ResNet-50 and YOLOv8 proceeded to the full evaluation.

5.3 Training Procedure

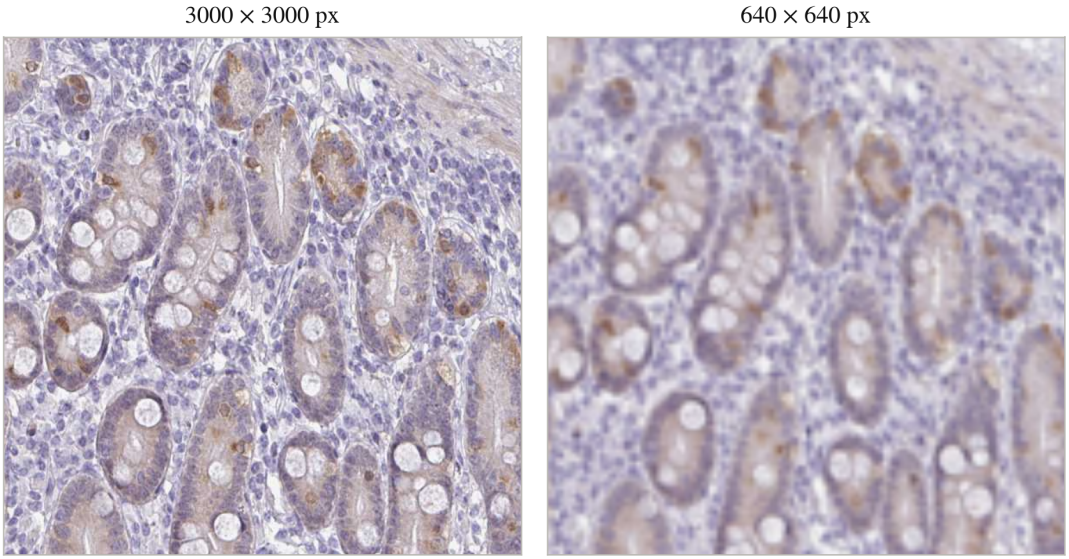

Figure 5-3: The same region of one tissue section at the scanner's native resolution and at the network's input size. The image is from the Human Protein Atlas (Uhlén et al. 2015).

Training followed a single recipe that applied identically across all folds. ResNet-50 uses ImageNet weights, and every layer is therefore updated during training. The gold dataset cannot support learning 25.6 million parameters from scratch. Each tissue section is resized to 640 by 640 pixels and normalised to the colour statistics the pretrained weights expect. The original scans are about 3000 pixels wide, so any practical input size inevitably loses detail, particularly in the fine features essential for labelling. Figure 5-3 shows the same region of one section at both sizes. At the native resolution, individual cells and their surfaces remain readable, while at 640 pixels the same structures blur together. Determining whether five cells align with both visible surfaces depends on this level of detail. The input size was therefore kept as large as the hardware permitted. Training ran on the university's shared GPU server, Garlick, where 8 GB cards hold a batch of six images at 640 pixels. That set the working resolution.

Training occurs in epochs, each representing a complete pass through the training data. After every epoch, the model is evaluated on the validation slice. Training ends if validation macro-F1 does not improve for six consecutive epochs, at which point the model restores the weights from the best epoch. This early stopping mechanism prevents overfitting by halting training once general learning has finalised. Optimisation uses AdamW (Loshchilov and Hutter 2019) with a learning rate of 10^{-4}, which determines the step size for weight updates. Training employs mixed-precision arithmetic. Intermediate values are stored at half precision, halving memory requirements per image, while weight updates retain full precision. This approach enables a batch of six 640-pixel images to fit within the available GPU memory.

The loss function decides what the network learns. It scores how wrong each prediction is, and training adjusts the weights to lower that score. The standard choice for classification is cross-entropy, which penalises the model in proportion to how little probability it assigned to the true class. To address the imbalance described in Section 5.1, each class's contribution to the loss is weighted,

$$\ell(x) = -w_y \log p_y(x), \qquad w_c = \frac{N}{2\,N_c} \tag{5.2}$$

Where:
$p_y(x)$ is the probability the model assigns to the true class y of image x

N is the number of training images
N_c is the number of training images belonging to class c

With 81 per cent of images usable, an error on an unusable image costs about four times as much as an error on a usable one, so the network cannot lower its loss by ignoring the rare class (Buda et al. 2018).

Augmentation expands the training data with label-preserving variations of each image. Every training image is randomly flipped and rotated by a multiple of 90 degrees, and its brightness and contrast are varied by at most ten per cent. Stronger photometric changes, meaning changes to the pixel values themselves rather than to the geometry, are deliberately excluded. Blur, noise and compression artefacts are the very defects a quality model must detect, so injecting them during training would teach the network to ignore its own signal (Schömig-Markiefka et al. 2021).

Two further common techniques were rejected. Focal loss concentrates learning on the examples the model finds hardest (Lin et al. 2017), but when some training labels are wrong, the hardest examples are disproportionately the mislabelled ones, so the loss would focus on exactly the labels least worth learning. Label smoothing replaces each hard target of 0 or 1 with a slightly softened value, so the model is never asked to be fully certain. It showed no measured benefit on the project's data and was dropped.

Machine labels can enter training in two ways. Naive pooling adds every machine label as a hard verdict and leaves the recipe unchanged. The alternative, hereafter the noise-aware recipe, adds three defences against label noise. First, each machine label carries its panel votes, letting the target reflect how contested the verdict was. An image settled unanimously at the first stage keeps a hard target of 1 or 0. An image escalated to the full panel, where all seven votes exist, receives a softened target.

$$\tilde{y} = [\,1 - f,\ f\,], \qquad f = \frac{v}{7} \tag{5.3}$$

Where:
v is the number of panel members who voted usable
f is the target probability for the usable class

The loss is then computed against this pair, rather than a hard label. This

approach softens supervision: a unanimous image contributes a strong training signal, while an escalated image never does.

At least one first-stage dissenter remains on the full panel, so the most lopsided vote an escalated image can receive is six to one. The closer the vote is to a tie, the closer the target is to 0.5 and the weaker the gradient is. As a result, the least certain labels exert the least pull on the network.

Second, the machine pool consists mostly of usable slides, so adding it shrinks the minority share from nineteen per cent to as little as six per cent in the trust-filtered configuration. At that distribution, the weight in Equation 5.2 rises above eight, and training becomes unstable: validation performance peaks in the first epoch and declines thereafter. The weighting was therefore replaced with balanced resampling, supported by a cosine schedule. Instead of penalising errors on rare images more heavily, the data loader draws them more frequently, so batches are nearly balanced and the plain loss suffices. This allows validation to settle into a stable plateau. This repair makes the two training sets comparable. Once optimisation is stable, any difference between the resulting models reflects the labels they learned from, not the formula they were trained with.

5.4 Training Data Selection

Table 5.2: Every contender under the shared protocol, judged on the 847 gold labels at both operating points. Gold only uses the expert labels alone. Naive adds every machine label as a hard verdict. Noise-aware softens each added label by its panel vote, balances the batches by resampling, and ends with a fine-tune on the gold labels alone. The confident subset keeps only near-unanimous machine labels. The trust-filtered set additionally discards the panel-voted unusable calls.

Architecture	Training data	Macro-F1		AUC	Unusable
		argmax	tuned		recall
ResNet-50	gold only (847)	0.8041 ± 0.0207	0.8199 ± 0.0506	0.9159	0.8007
YOLOv8s-cls	gold only (847)	0.7758 ± 0.0615	0.8100 ± 0.0669	0.9113	0.6564
ResNet-50	combined, naive (847 + 5,141)	0.6985 ± 0.0337	0.7503 ± 0.0851	0.8886	0.9184
ResNet-50	combined, noise-aware (847 + 5,141)	0.7583 ± 0.0490	0.7626 ± 0.0240	0.9029	0.8361
ResNet-50	combined, confident subset (847 + 4,249)	0.7514 ± 0.0749	0.7609 ± 0.1069	0.8993	0.8072
ResNet-50	combined, trust-filtered (847 + 4,834)	0.7775 ± 0.0448	0.7878 ± 0.0599	0.9059	0.7734
YOLOv8s-cls	combined, trust-filtered (847 + 4,834)	0.8152 ± 0.0377	0.8159 ± 0.0570	0.9082	0.7509

With the architecture and training protocol fixed, the effect of increasing training data could be assessed. Expanding the gold set from 64 to 163 unusable images raised unusable recall from 0.59 to 0.672 and reduced fold-to-fold variation from ± 0.28 to ± 0.18, demonstrating that additional labels enhanced model performance. The machine-labelled pool enabled further scaling, contributing 5,141 additional labels. Both architectures were trained using only gold labels and with combined datasets that included machine-labelled images. Moreover, Table 5.2 lists each model by its architecture and training data. All other aspects of the protocol were held constant. Meaning, any observed differences between models were attributable solely to the training data.

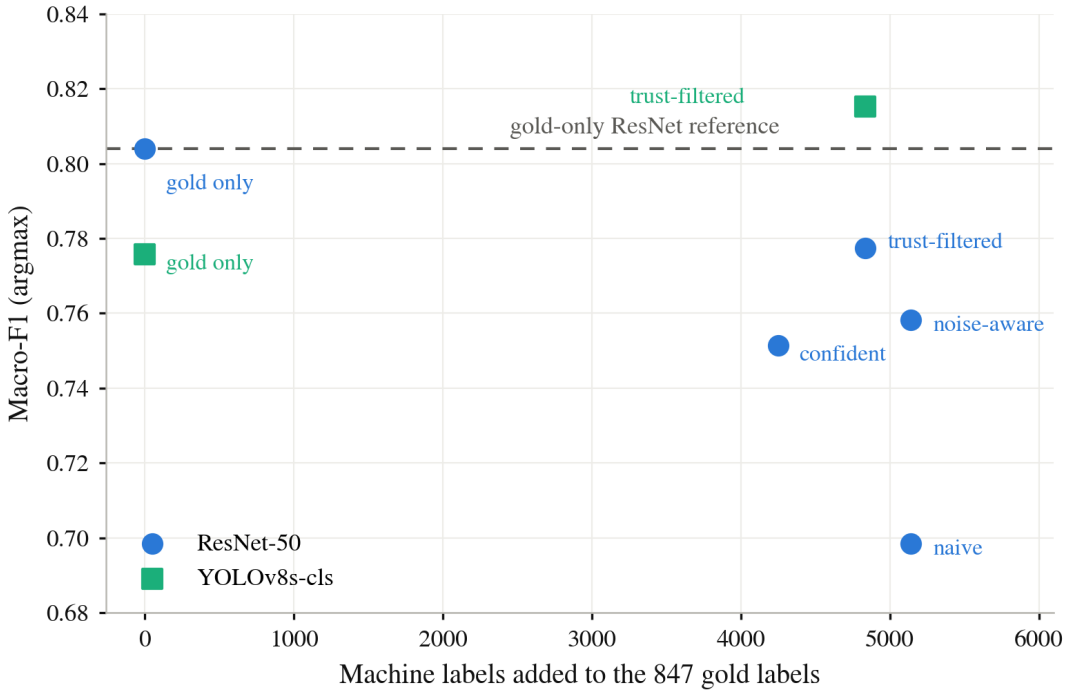

Figure 5-4: A scatter plot of argmax macro-F1 as machine labels are added, for both architectures. The combined treatments span 0.6985 to 0.8152, against the gold-only ResNet reference of 0.8041 marked by the dashed line.

As Figure 5-4 exhibits, the expansion never paid off for the ResNet. Naive pooling lost an average of 0.106 in macro-F1 and fell short on all five folds. The noise-aware recipe narrowed the average loss to 0.046. No combined treatment's aggregate crossed zero. Restricting the machine labels to the confident subset reached 0.7514. The 759 images with both an expert and a machine label explain the pattern, as they let the machine labels be checked directly against the expert. On this overlap, agreement ranges from 83.3 to 97.8 per cent across the machine labels' usable calls, but from only 33.8 to 72.1 per cent across their unusable calls, depending on how the panel voted. The added volume is therefore cleanest where the model is already strong and noisiest in the minority class it needs most. A final procedure enabled the trust-filtered set. This kept only the machine labels from the strata that agreed well and discarded the panel-voted unusable calls. However, at 640 pixels it reached 0.7775, the best combined result in Table 5.2. However, it was still below gold-only at both operating points.

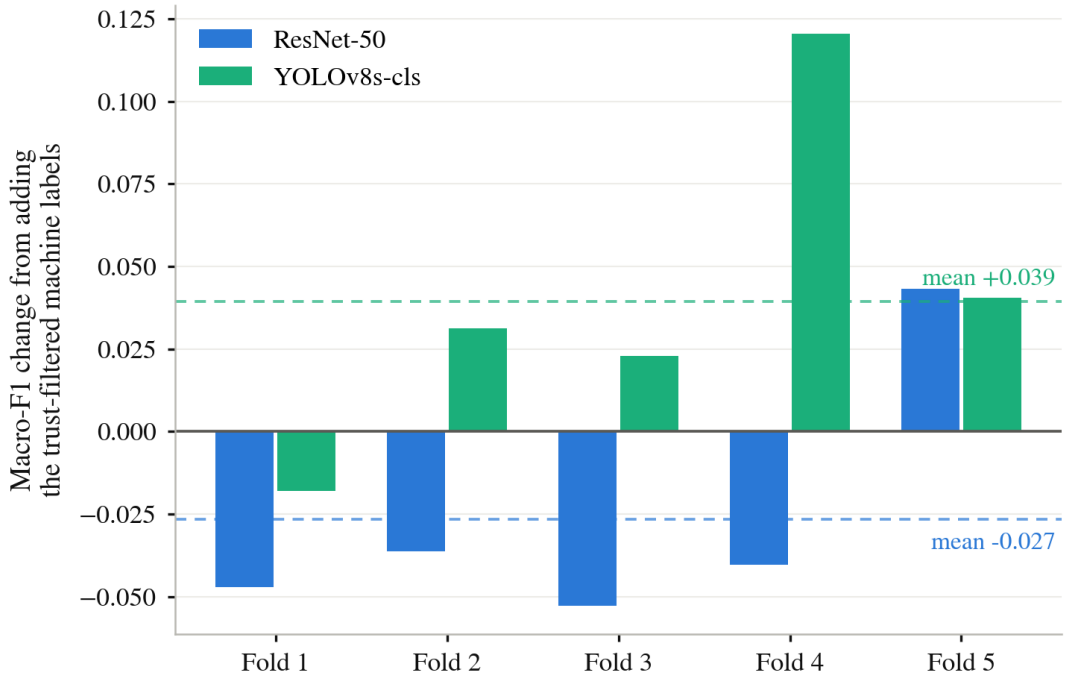

Figure 5-5: A bar chart of the per-fold change in macro-F1 from adding the trust-filtered machine labels, with mean changes of +0.039 for YOLOv8s-cls and −0.027 for the ResNet.

The second architecture produced a different outcome. Using the same trust-filtered data and folds, YOLO's performance improved by 0.039, whereas ResNet's declined by 0.027, as illustrated in Figure 5-5. Thus, the same labels benefited YOLO but hindered ResNet. This difference likely arises because the YOLO network rarely encounters the same image view. Hence, a wrongly labelled image appears only in fragments, making its label harder to memorise.

Additionally, the running average of YOLO's weights helps smooth out sharp updates caused by individual incorrect labels. In contrast, ResNet always trains on the entire section, meaning a wrong label is repeated with nearly identical evidence in every epoch. While these mechanisms are plausible explanations, the experiment compared the systems as complete units rather than isolating individual components, so the precise cause remains unclear.

Therefore, the decision for the ResNet was straightforward. The model taken forward is the one trained on gold labels alone. Each run in this comparison used a single fixed random seed, so the conclusions reflect that constraint. Throughout

the study, performance was never limited by the amount of training data but by label quality.

5.5 The Final Model

As shown in Table 5.2, the gold-only ResNet achieved 0.8041 ± 0.0207 at argmax and 0.8199 ± 0.0506 when tuned. In addition, the combined-data YOLO reached 0.8152 ± 0.0377 and 0.8159 ± 0.0570, respectively. At both operating points, the performance difference fell within the margin of fold-to-fold variation. This resulted in a statistical stalemate. An exact McNemar's test on the 847 paired test predictions supports this reading, with no significant difference at either operating point ($p = 0.271$ at argmax, $p = 0.461$ tuned). Yet, the two models differed in consistency. When trained solely on gold data, YOLO dropped to 0.7758 with three times ResNet's fold variance, indicating its reliance on machine labels. ResNet showed greater stability, with about half the fold spread of YOLO and more consistently tuned thresholds. For example, ResNet had 0.29–0.54 compared to YOLO's 0.19–0.74. Based on these results, ResNet-50 was selected as the production model. Nevertheless, YOLO played a crucial role in uncovering the asymmetric quality of the machine-generated labels through its contrasting performance.

Chapter 6

Localising The Analysable Regions

6.1 Task, Supervision And Method

After establishing the screening process, the next objective is to identify the readable epithelium within each usable section. Without tile-level training annotations, supervision relies exclusively on image-level labels. This challenge is addressed through multiple-instance learning (Campanella et al. 2019), in which each section is represented as a bag of 64 tiles. The model must infer which tiles contribute to the overall verdict without explicit tile-level guidance.

Each tile is embedded using the Stage-1 ResNet-50 encoder at 448 pixels, an input size close to the tiles' native 375-pixel width. These encoders are fold-matched, such that each fold's tiles are embedded by a checkpoint trained without that fold's donors. Employing a single common encoder would risk introducing data leakage from training donors into other folds' test sets. A gated attention head subsequently aggregates each bag into an image-level prediction (Ilse et al. 2018). Localisation is derived from this prediction structure,

$$z = \sum_{i=1}^{64} a_i \, \psi(h_i), \qquad c_i = a_i \, \psi(h_i) \tag{6.1}$$

Where:
h_i is the learned representation of tile i
a_i is the attention weight of tile i, normalised over the section
$\psi(h_i)$ is the tile's individual verdict

The contribution c_i is therefore the tile's exact share of the image-level prediction. Tiles are ranked by contribution rather than by attention, since an attention weight records where the model looked while a contribution records how much a tile moved the verdict (Javed et al. 2022). Because the prediction is the sum of its contributions, the resulting map is the decomposition of the verdict itself rather than a post-hoc explanation.

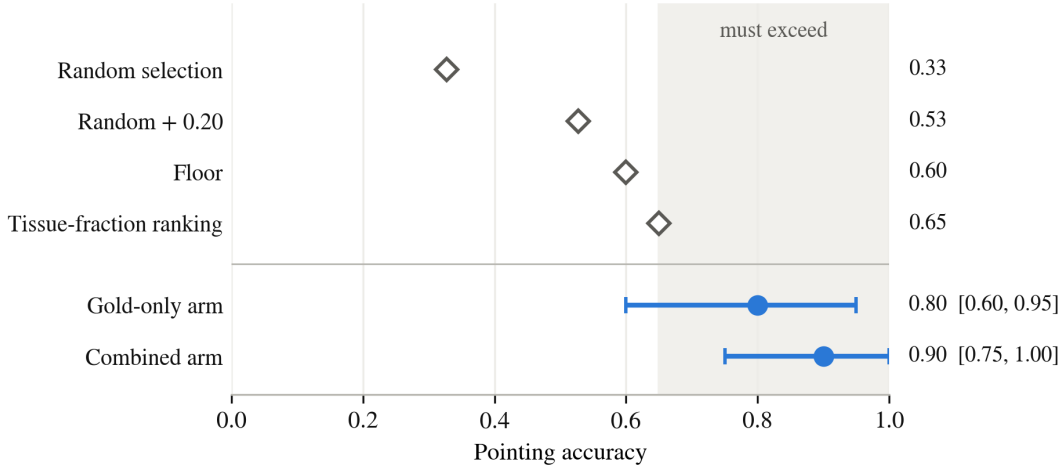

Figure 6-1: A dot-and-interval chart of pointing accuracy for the two localiser arms, 0.80 for the gold-only and 0.90 for the combined, against the three requirements fixed before evaluation, with 95 per cent confidence intervals from section-clustered bootstrap resampling. The shaded region begins at the strictest requirement, the ranking of tiles by tissue fraction at 0.65.

Within each fold, all tuning is confined to the validation set. The localiser must satisfy two tests. The first is parity. Adding localisation must not reduce image-level accuracy. Specifically, the mean paired-fold delta in macro-F1 against the 448-pixel gold-only ResNet-50, both at argmax, must remain above -0.02, with no fold falling below -0.10. Moreover, the second test is pointing, which requires that the model's highest-scored tile correspond to one identified as qualifying by the domain expert (Zhang et al. 2018). Pointing accuracy is defined as the proportion of sections for which this criterion is met. It must meet three objectives: achieve at least 0.60, exceed random selection by 0.20, and surpass a ranking based on tissue fraction. Figure 6-1 presents these three objectives on a unified scale. The strictest criterion is placed at 0.65. As secondary measures,

top-three and top-five accuracy count a success whenever a qualifying tile appears among the model's three or five highest-scored tiles.

6.2 The Expert Tile Ground Truth

Localisation required region annotations, which had not been created previously. The domain expert evaluated a complete tile map for 20 usable sections using the same annotation platform as in Stage 1. In total, 1,280 tiles were reviewed. Of these, 834 were assessed directly, with 419 marked as qualifying and 415 as non-qualifying. The remaining tiles were classified as background and recorded as non-qualifying. In summary, 419 out of 1,280 tiles qualified, and every section contained at least one. This set served evaluation only and was never used in training or tuning.

6.3 Results

Trained on the 847 expert-labelled images alone, the gold-only MIL achieves a pointing accuracy of 0.80, as shown in Figure 6-1, clearing all three requirements. Random selection reaches 0.33 and the tissue-fraction ranking 0.65, so both baselines are passed with the required margins. Top-three and top-five accuracy reach 0.95. The contribution scoring is justified because ranking the same model's tiles by raw attention points correctly in only 0.65 of sections. Parity, however, fails because the mean delta is -0.024 against the -0.02 bar, with the worst fold at -0.157. Therefore, localisation traded roughly two macro-F1 points of image-level accuracy for region-level evidence.

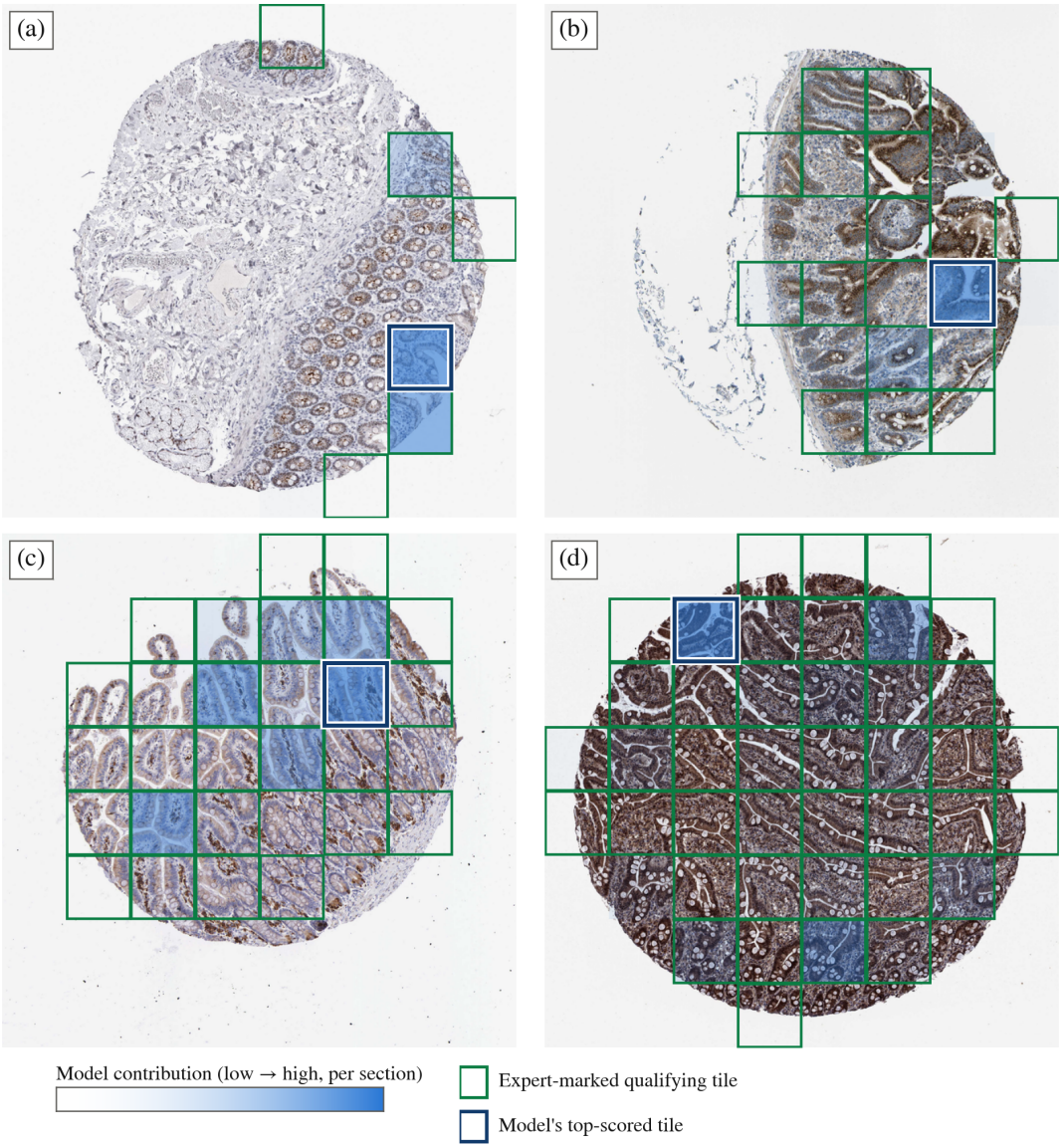

Figure 6-2: Contribution maps of the combined-data MIL over four gold-evaluation sections, from sparse (a) to abundant (d) qualifying tissue. Green outlines mark the expert's qualifying tiles, the boxed tile is the model's highest-scored, and the images are from the Human Protein Atlas (Uhlén et al. 2015).

To test whether additional supervision helps this regime, the 5,141 machine-labelled images of the combined dataset were joined to training as further weak bags. Pointing accuracy rises to 0.90, with top-three and top-five accuracy both reaching 1.00. As Figure 6-1 shows, only this arm's confidence interval lies entirely

above the strictest requirement. Parity now passes, with a mean delta of -0.009 and a worst fold of -0.039. No leakage supports these figures, since the machine bags exclude all 847 expert-labelled filenames and the expert tiles were never used for training. As a result, the same machine labels that never improved the screening classifier improve the localiser. A likely reason is that MIL copes well with many weakly labelled bags (Campanella et al. 2019). Figure 6-2 shows the result, with the model's highest-scored tiles falling inside the expert-marked stretches.

Chapter 7

Discussion And Conclusion

7.1 Evaluation of Achievements Relative to Objectives

This section evaluates the two-stage pipeline against its initial objectives to assess what the project has achieved. The first was met by a labelling rubric turning expert usability judgement into a written, repeatable standard. The second was met by showing the models train well on IHC images, notably under donor-grouped cross-validation. The third showed that expert-corrected machine annotations improved localisation, albeit at the slice classifier's expense. Beyond this, a functional localiser was implemented, capable of identifying an expert-confirmed qualifying stretch from image-level labels alone. That said, the quantitative performance target fell short. The ResNet50 model trained on the gold-standard dataset achieved a tuned macro-F1 score of 0.82, compared to the objective of 0.85. This target was set below a known upper limit; with a kappa of 0.735, a model trained on these labels cannot realistically exceed the expert annotator's own consistency. Current evidence does not establish whether a more advanced model could close the remaining performance gap. Beyond its stated objectives, the project's most durable contribution may be the datasets it leaves behind: 847 expert-labelled images and a 1,280-tile localisation dataset, the first such resources for IHC intestinal tissue. Future work no longer needs to start from zero.

7.2 Limitations And Future Work

The project's chief limitations stem from the training procedure. Machine labels improved YOLO's performance but reduced ResNet-50's; this raises the question of why identical labels produce opposite effects. Since the two systems differ in several components, isolating each one would be necessary to identify the underlying cause. A further limitation is the use of a single random seed. All training runs in this project used the same fixed seed, so variation between folds is quantified, but variation between repeated runs is not. Repeating the final comparison across multiple seeds would help confirm whether the conclusions hold.

Beyond addressing seed variation, expanding the validated dataset with additional expert annotation would improve both model and localisation performance. The localisation stage received a narrower investigation than the classification stage. Stage 2 evaluated only a single localiser architecture under two training sets. Whether alternative tile grids, aggregation heads, or encoders would improve localisation remains untested. The expert tile set was reserved for evaluation, so no model has been trained directly on region labels. An expanded tile set would enable this stronger form of supervision while preserving the current set for assessment. More broadly, cataloguing the spatial distribution of proteins across the intestinal epithelium requires reading each protein's compartment from the images. Since the localiser provides this input, the next step in developing the pipeline would be compartmental reading.

7.3 Conclusion

Each examination of a tissue section begins with an implicit judgement: whether the section is readable. The Human Protein Atlas has never formally recorded this assessment (The Human Protein Atlas 2022). Consequently, histologists must re-evaluate protein localisation before any biological analysis. The pipeline is only the start of a more comprehensive system that could catalogue proteins to assist histologists in reviewing tissue samples. The slice quality classifier, alongside the localiser that guides histologists, represents a modest yet meaningful step towards more efficient research in this area.

Code And Data Availability

The source code, configuration, and result artefacts supporting this dissertation are held in a repository at `https://github.com/Thinuka112/diss`.

Appendix A

Annotation Platform User And Maintenance Guide

This appendix records how the annotation platform described in Chapter 4 is operated, deployed and maintained. The platform is a FastAPI web service (Ramírez 2018) with a single-page browser client built on version 4.1.0 of the OpenSeadragon viewer (OpenSeadragon contributors 2025). The image collection is packaged inside the application's container, while every judgement is written to persistent storage outside it.

A.1 Signing In And Judging

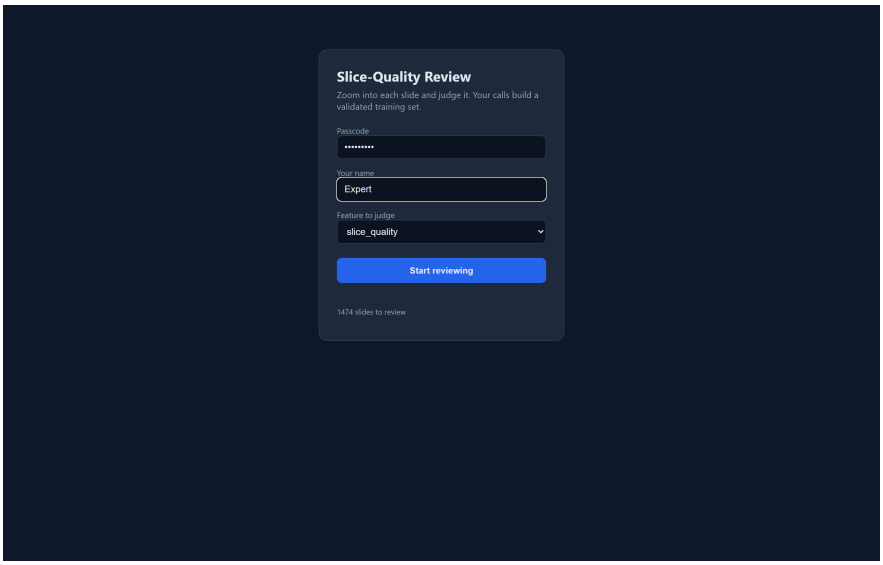

Figure A-1: The platform's sign-in view. The reviewer enters the shared passcode, a name on the allowlist and the feature to judge.

The reviewer opens the platform's web address in any browser and signs in with the shared passcode, a name and the feature to judge, as shown in Figure A-1. The name must match an entry on the reviewer allowlist, and the feature selects the judging task, such as `slice_quality` for the whole-image verdicts or the tile task used in the localisation campaign. The judging view then serves one image at a time, and a single keypress records each verdict and advances the queue. An undo key reverses an accidental judgement. The browser can be closed at any point without loss, as every verdict is stored on the server the moment it is pressed, and the queue restores the reviewer's position on return.

A.2 Data Storage And Recovery

Judgements are stored in an append-only event log. Each sign-in opens a session file named `session_{feature}_{reviewer}_{timestamp}.csv`, and every keypress appends one row carrying a microsecond timestamp, the image identifier, the action and the verdict. No row is ever rewritten; an undo appends a reversal row, and the current label of an image is resolved on read as its most re-

cent event. The session directory resides on a persistent volume that is mounted into the container, so the judgement record survives restarts, redeployments and crashes. Recovering the platform therefore requires only redeploying the container and reattaching the volume.

A.3 Deployment And Maintenance

The platform is packaged as a Docker container on a slim Python 3.12 image, served by uvicorn on port 8000, and deployed on Fly.io in the London region. It runs as a single machine, since the file-based store is not safe across multiple instances, and the machine stops automatically when idle and restarts on the next visit within a few seconds. Four secrets are set at deployment and are never stored in the repository: the shared passcode, the administrator token, the cookie-signing secret and the reviewer allowlist. Redeployment is a single `fly deploy` command; the cookie-signing secret must remain stable across redeployments or sign-in sessions reset.

A.4 Exporting The Labels

Two administrative endpoints are gated by the administrator token. The export endpoint returns the resolved dataset as a comma-separated or spreadsheet file keyed on the image filename, which joins directly to the pipeline tables of Chapter 3, and the statistics endpoint reports review progress. Exports read the event log at request time, so a download always reflects every judgement made up to that moment.

Appendix B

The Usability Rubric

This appendix reproduces the operational rubric that Chapter 4 summarises, as the domain expert applied it across the labelling campaign. It is recorded here so that any future assessor, human or automated, can apply the same criterion to new slides.

B.1 Purpose And Unit Of Judgement

The rubric decides whether a slide can serve the downstream protein-distribution analysis, which depends on epithelium that reads along the apical-to-basal axis. Each judgement covers one full HPA image at native resolution, zooming wherever detail demands it, and assigns a single verdict of usable or unusable.

B.2 The Decision Rule

A slide is usable when it contains at least one qualifying stretch anywhere on the slide, at any orientation. The state of the remaining tissue carries no weight, so poor or complex regions elsewhere never disqualify a slide that holds one readable stretch. When no qualifying stretch is present, the slide is unusable.

B.3 The Qualifying Stretch

A qualifying stretch must satisfy all three of the following conditions.

1. At least five adjacent epithelial cells lie in a clear linear arrangement.
2. Both surfaces are visible, with the apical surface facing the lumen and the basal surface facing the underlying tissue.
3. The staining reads consistently along the whole stretch, so an apically expressed target must show consistent apical labelling from end to end.

The stain colour plays no part in the judgement, and the brown antibody signal and the blue nuclear counterstain are equally acceptable. The five-cell threshold exists to keep human judgement repeatable, while the classifier is expected to learn the visual pattern of a qualifying stretch rather than to count cells.

B.4 Indicators Of An Unusable Slide

Several patterns reliably signal that no qualifying stretch is present. Crypt cross-sections arise when the sectioning plane passes horizontally through the villus base, leaving circular and tightly packed cell profiles in which no five cells align. Regions of gland cells, and regions dominated by goblet cells, do not stain consistently enough to read. The transition zones between crypt and villus mix cells at different developmental stages, which makes any reading unreliable. When a slide remains genuinely ambiguous after full inspection, the default verdict is unusable.

Word Count

The main body of this dissertation contains 10,388 words. Following the programme's definition of the count, this figure covers the body text of Chapters 1 to 7 and excludes headings, captions, and the contents of figures and tables.

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